

Study of the entropy of two-qubit gates with uniform probability distribution using the minimal cut of circuits

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Minimal cut.



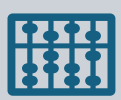
Graphs.



Karger's algorithm.



Random unitary circuits.

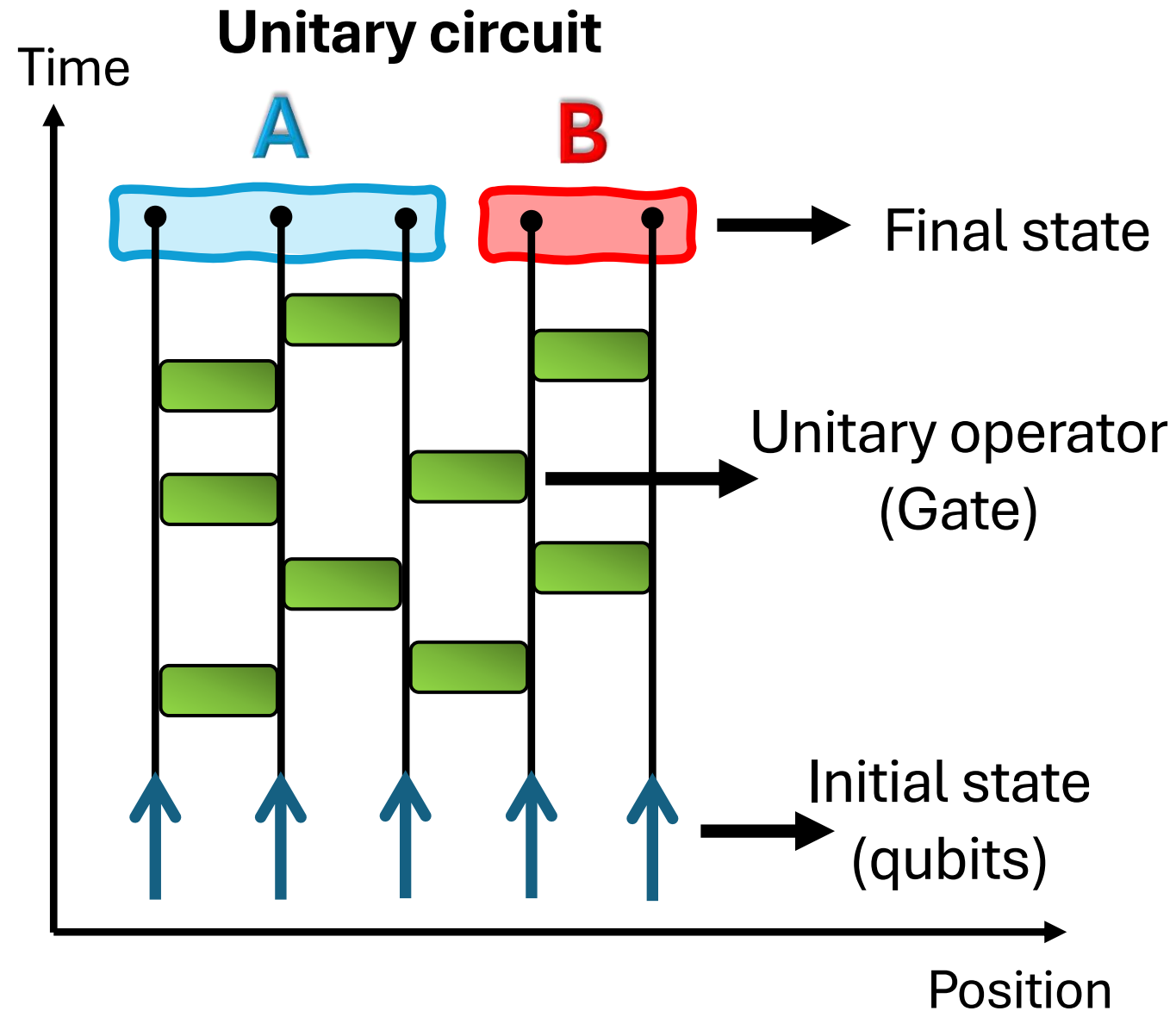


Study of two-qubits gate with uniform probability.



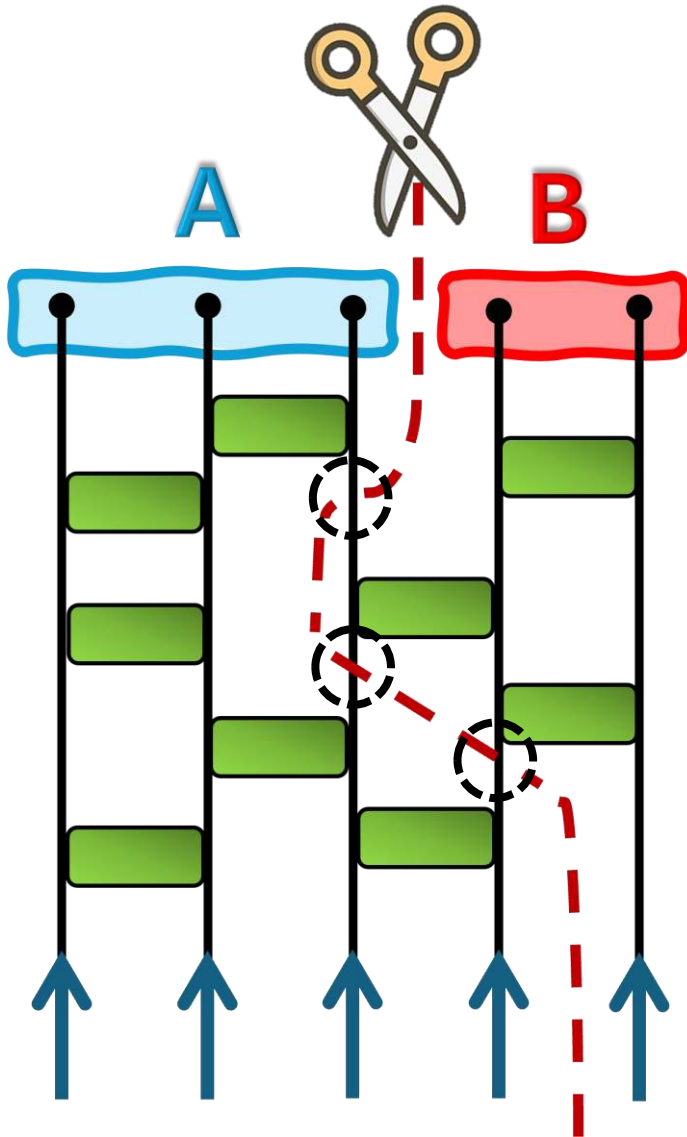
Conclusions.

MINIMAL CUT



To quantify entanglement in two quantum subsystems **A** and **B**, we use entropy.

MINIMAL CUT



Rényi entropy

$$S_n = \frac{1}{1-n} \log (\text{Tr}(\rho_A^n)) = \frac{1}{1-n} \log \left(\sum_i \lambda_i^n \right)$$

A **classical** upper bound for the entropy is achieved by minimal cut formalism

$$S_n \leq N_{\text{min cut}}$$

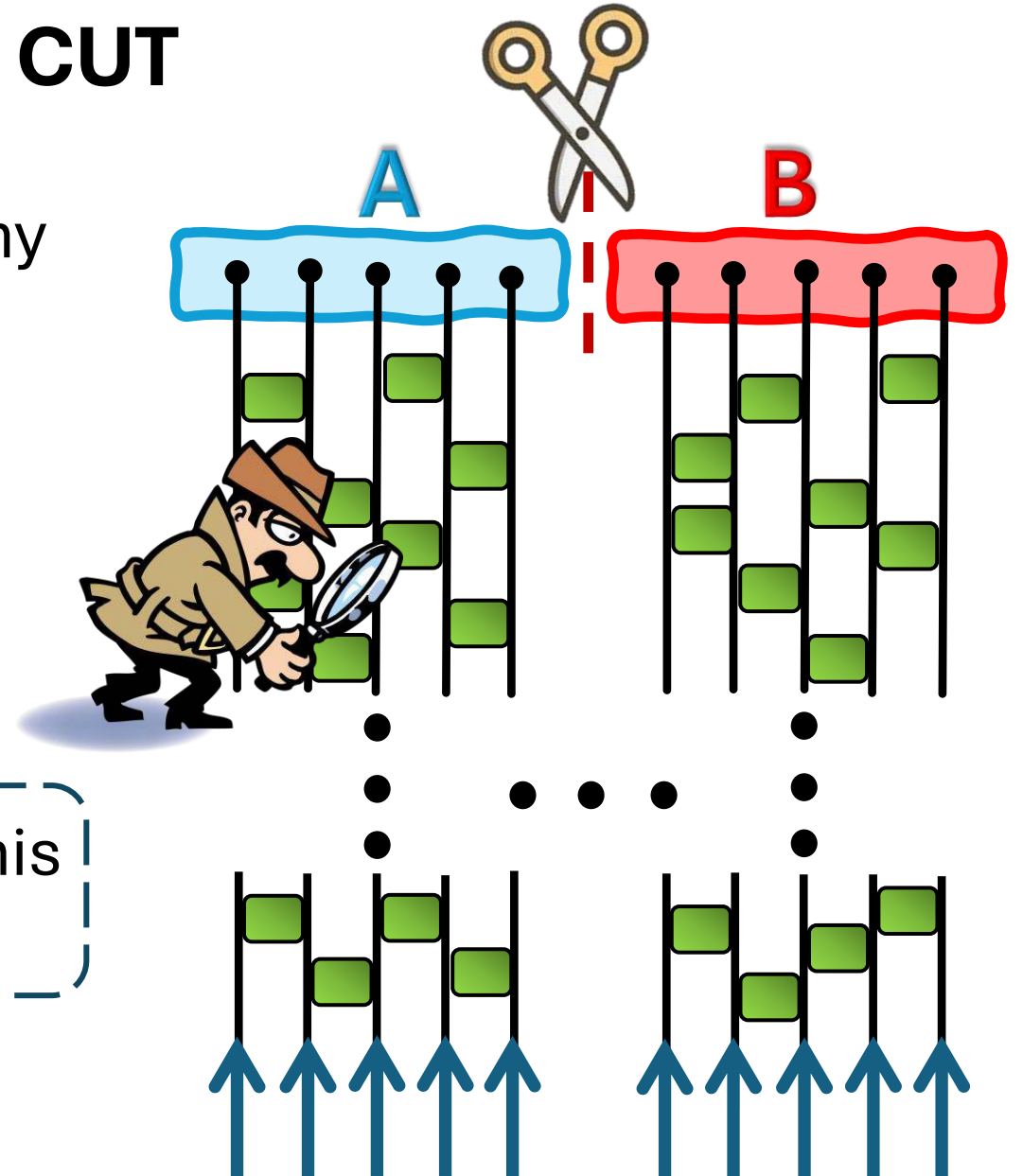
MINIMAL CUT

If the circuit is very complex, with many gates and qubits, it's not simple.

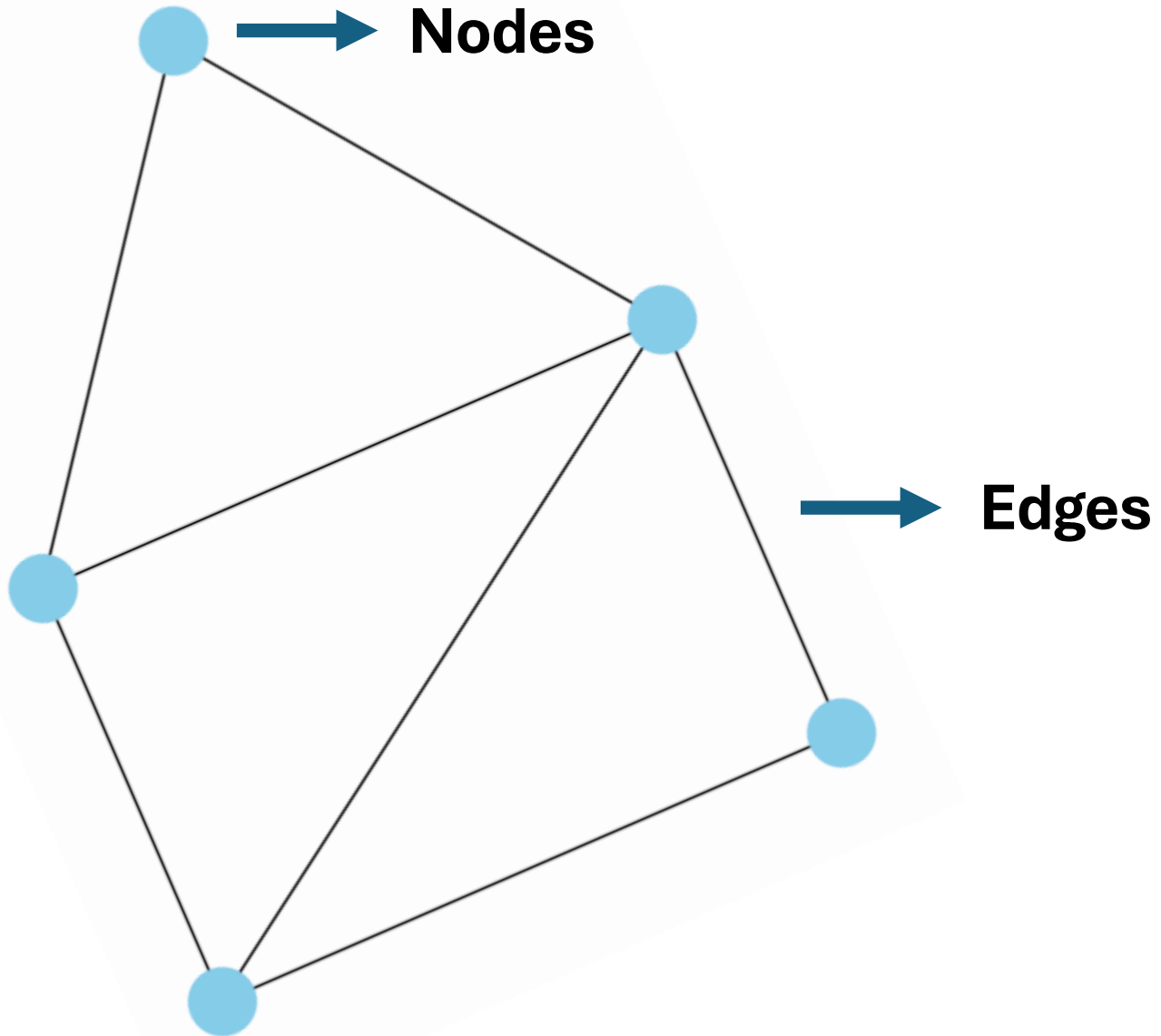
How to find the minimal cut?



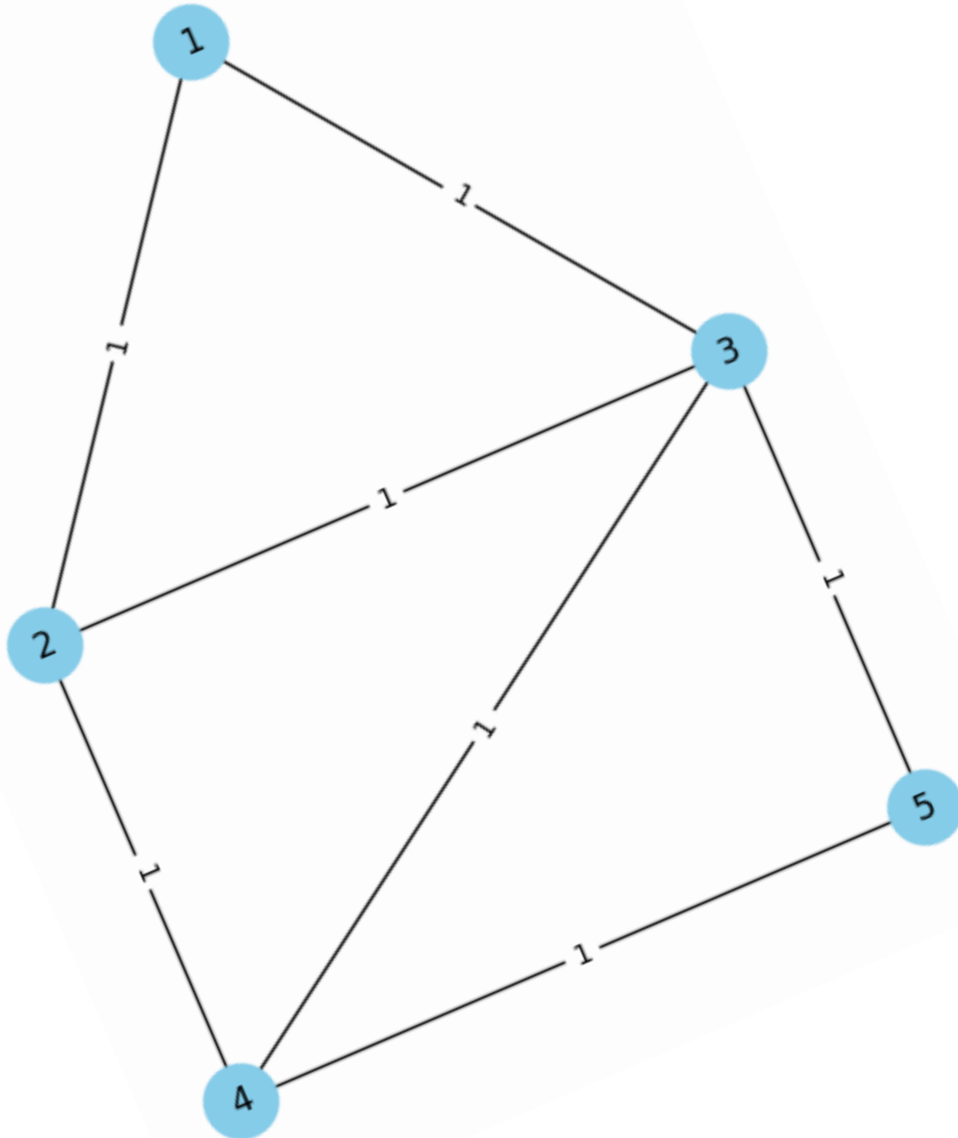
In graph theory there is a solution to this problem.



GRAPHS



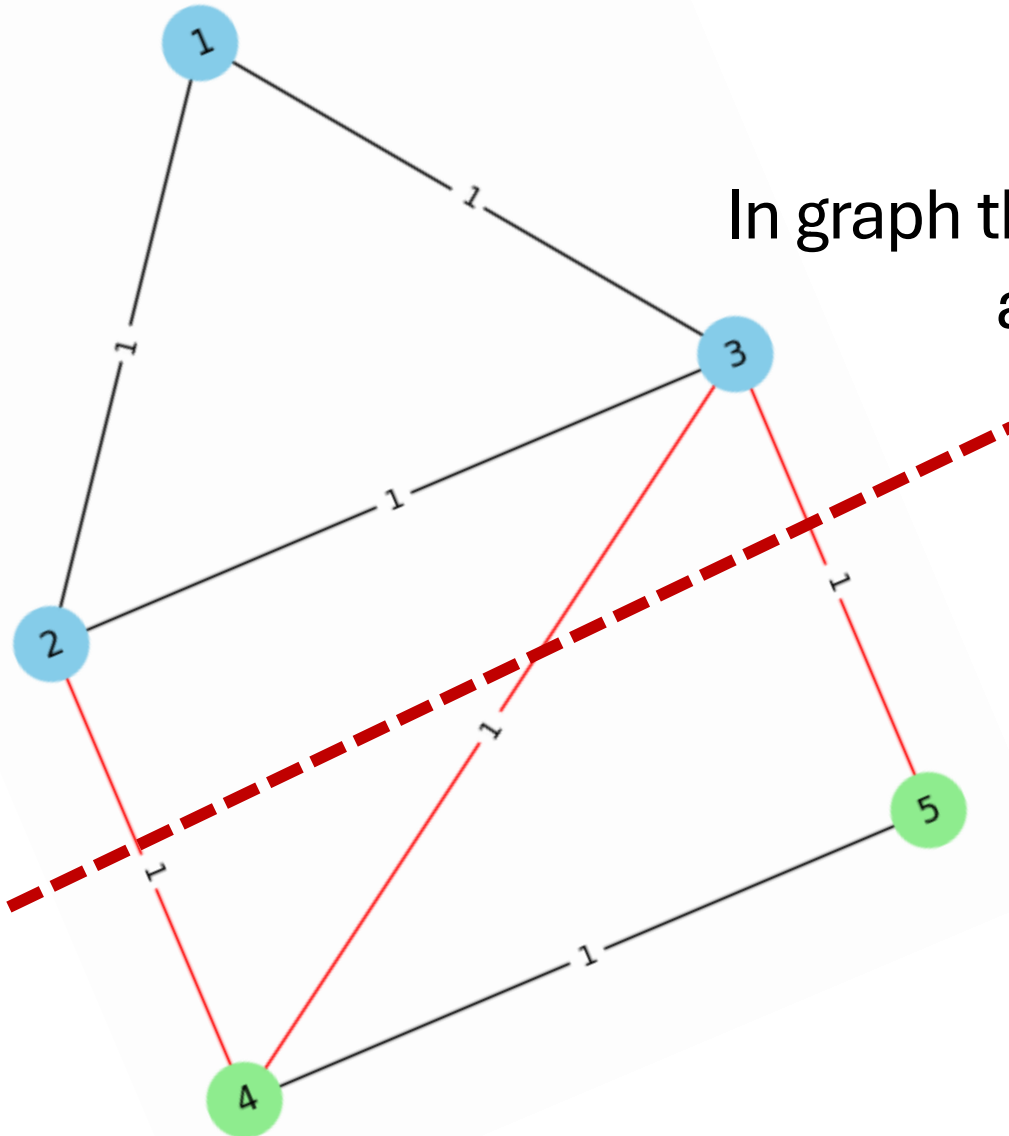
GRAPHS



- ❑ Nodes are labeled
- ❑ Weights can be assigned to edges

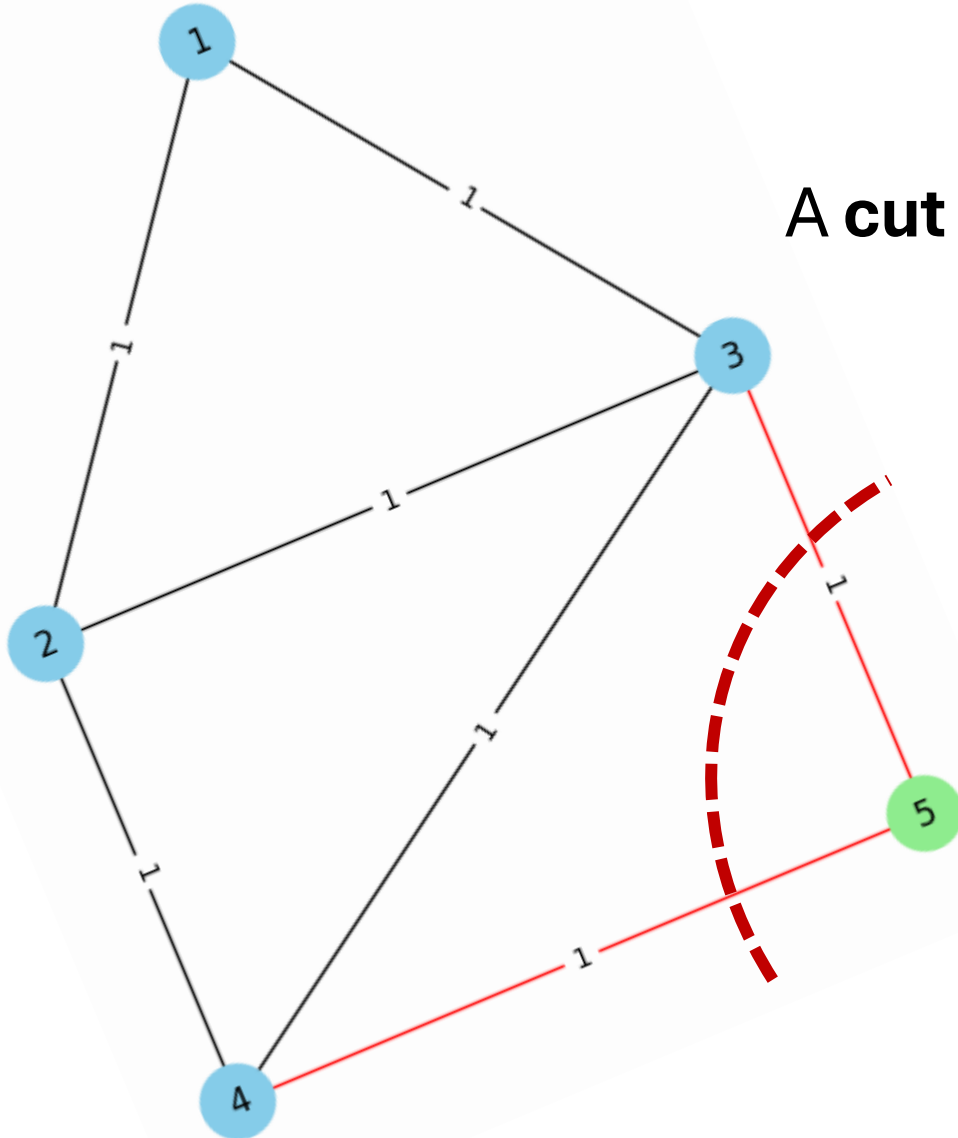
GRAPHS

In graph theory, a **cut** is a partition of the vertices of a graph into two disjoint subsets.

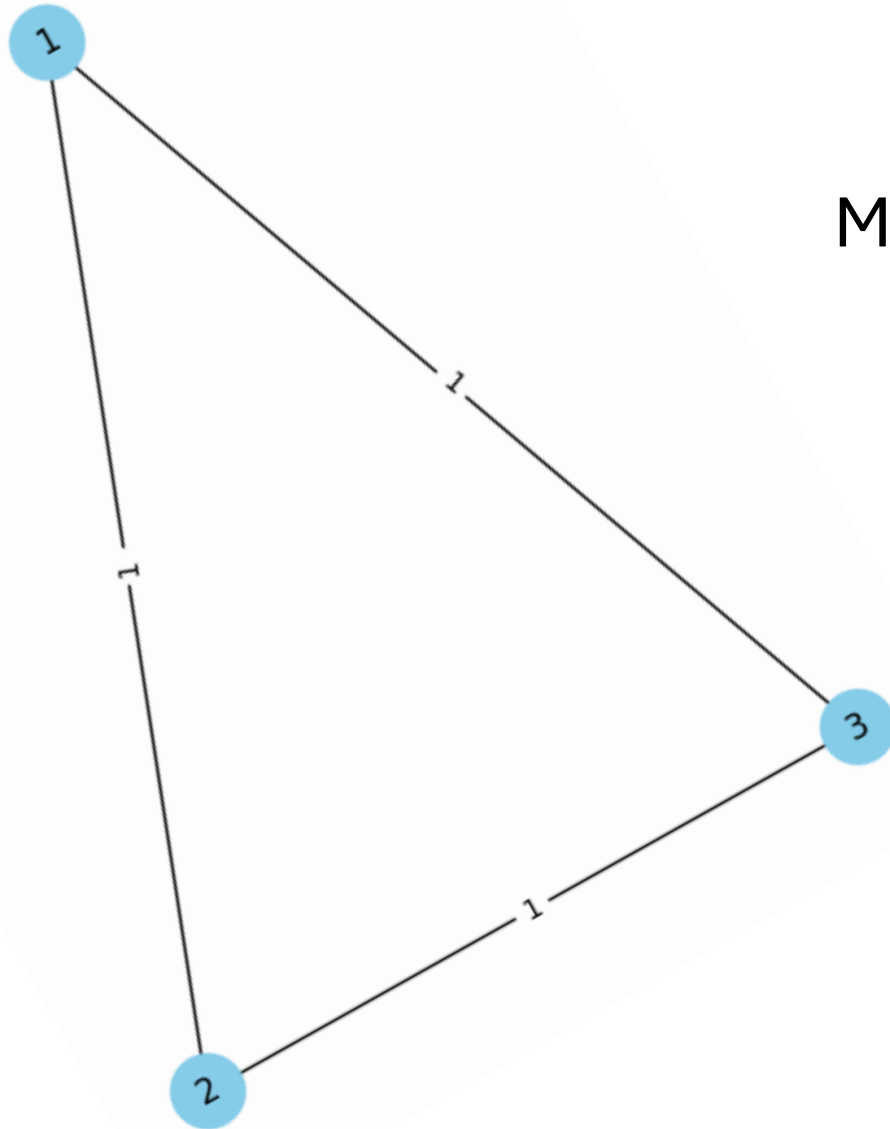


GRAPHS

A **cut** is **minimum** if the weight of the cut is not larger than the size of any other cut



GRAPHS



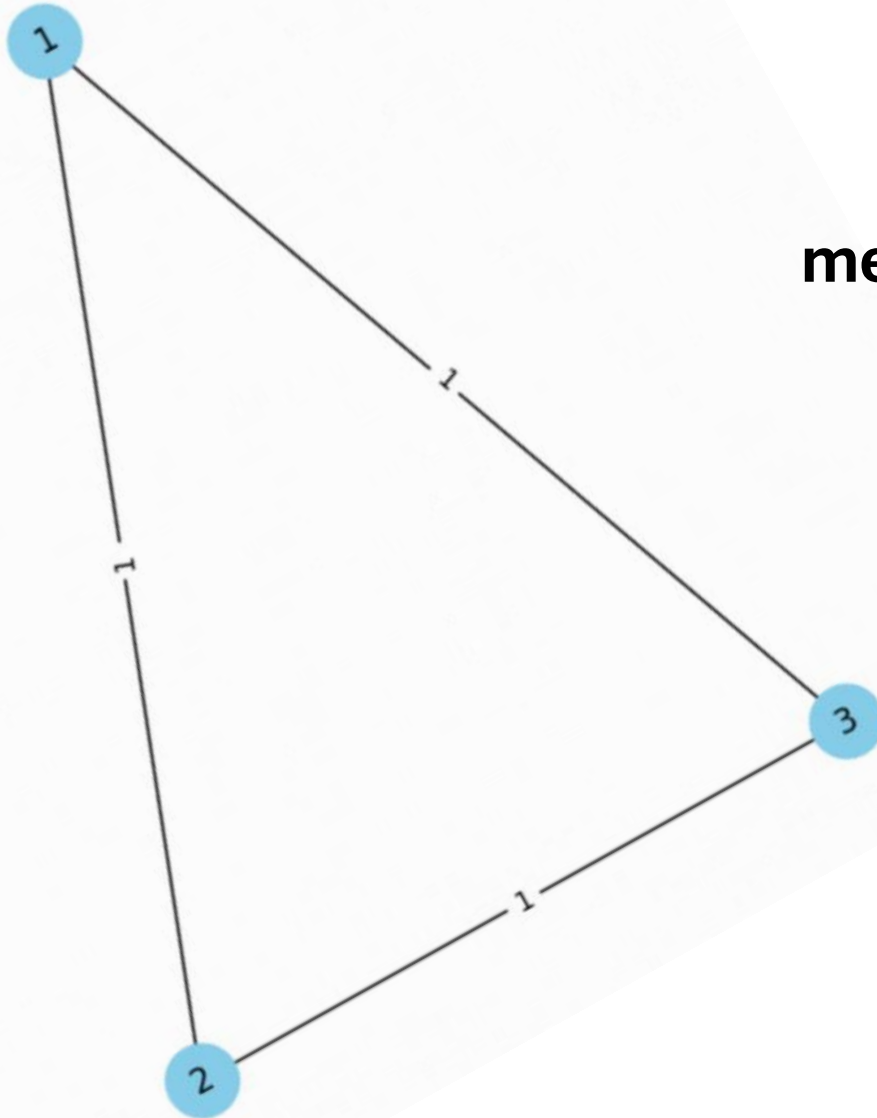
Minimum cut search algorithms use a simple
operation



Edge contraction (Merge nodes)

Is an operation that removes an edge from a graph while simultaneously merging the two vertices that it previously joined

GRAPHS



merge nodes 2 and 3



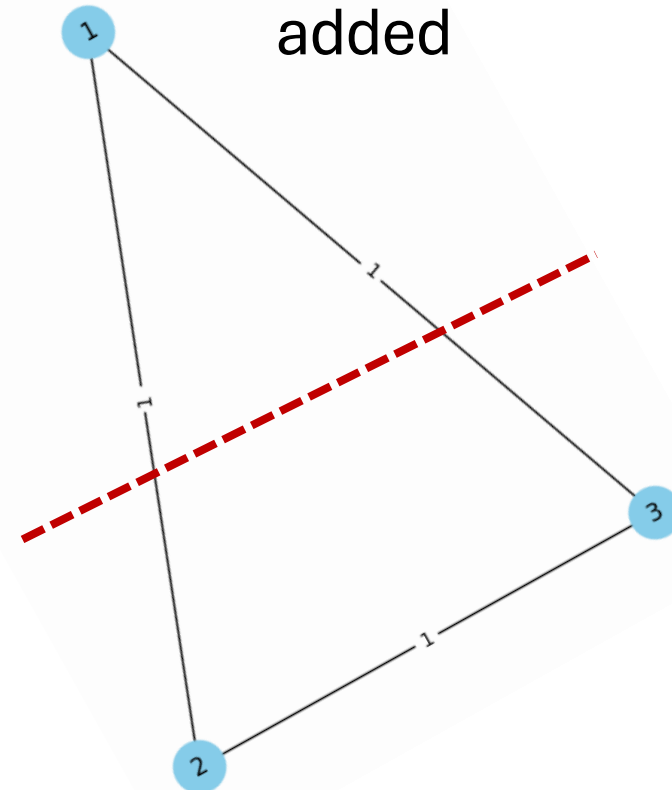
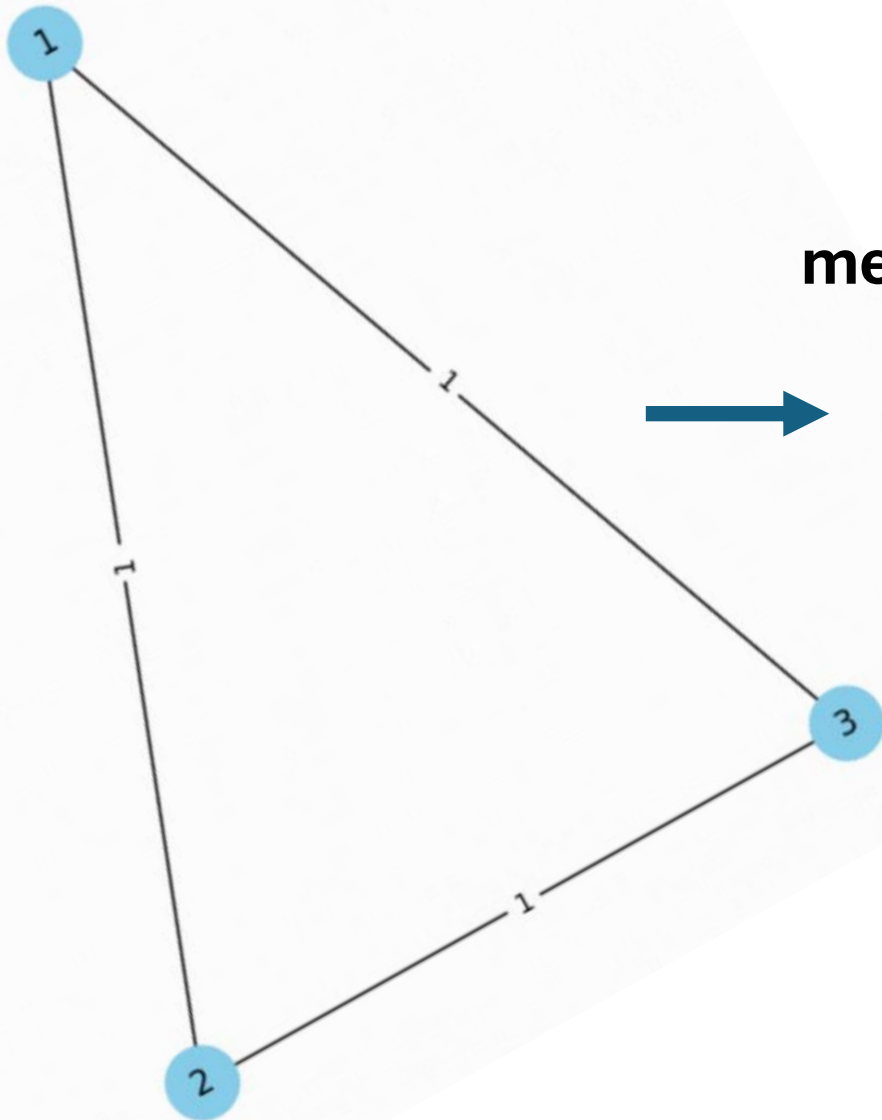
- ☐ The edge between nodes **2** and **3** disappears
- ☐ The edges that connect 2 and 3 with a common node are added.

GRAPHS

merge nodes 2 and 3



Represents the cut along the edges that were added



KARGER'S ALGORITHM

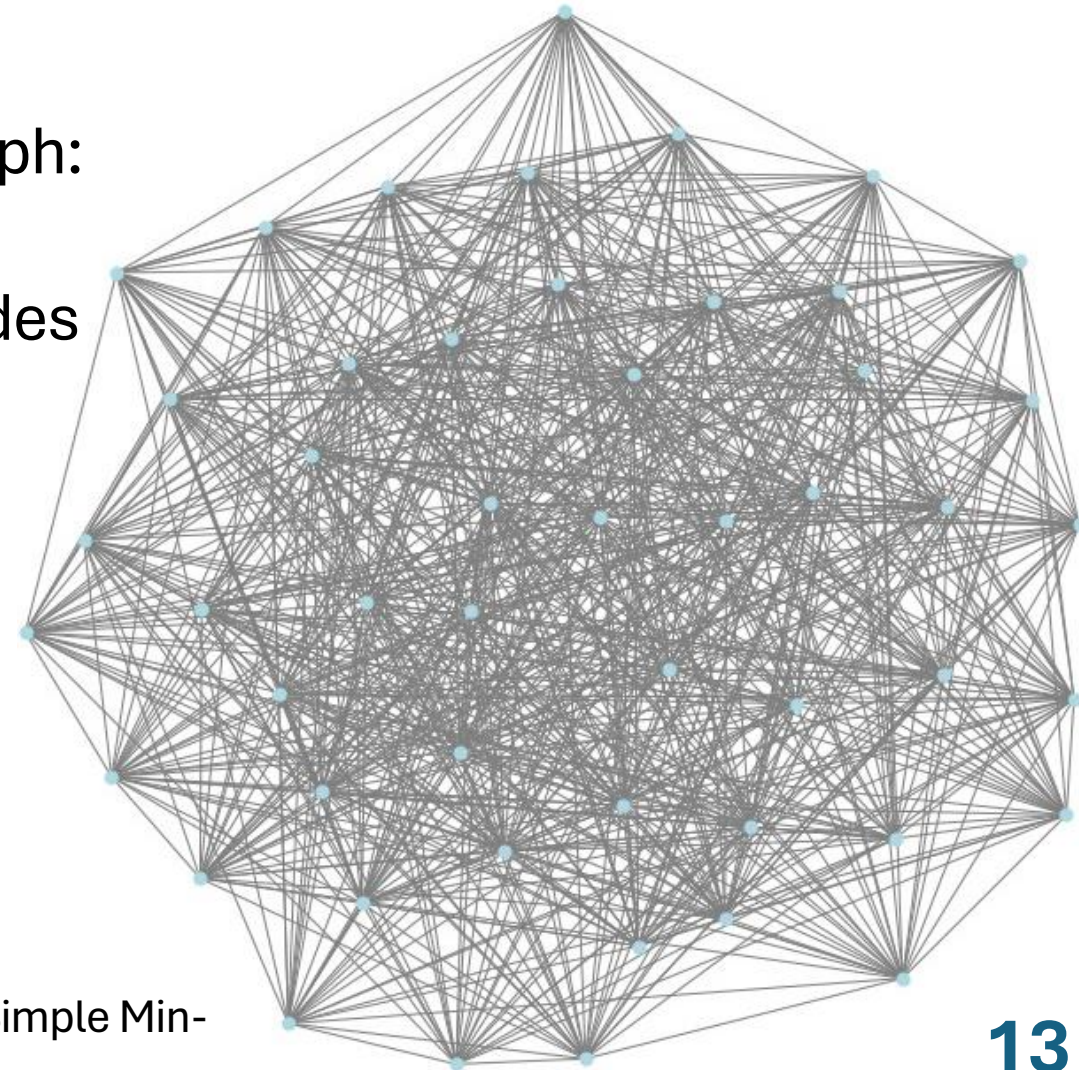
 It is a randomized algorithm to compute a **minimum cut** of a graph.

1. While there are more than two nodes in the graph:

1. Randomly select two adjacent nodes (nodes connected by an edge).
2. Merge the selected nodes into one.

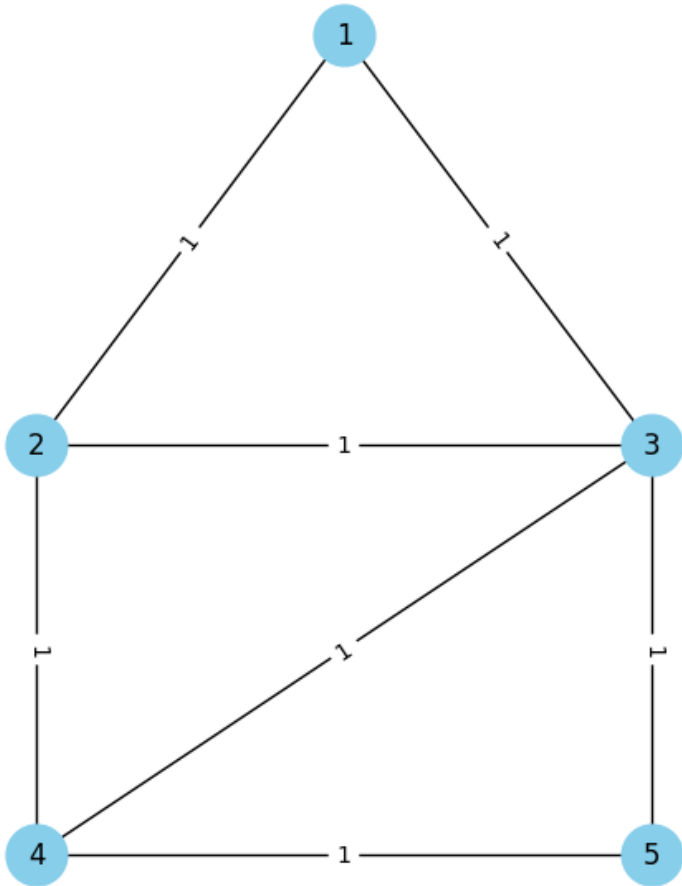
2. Return:

The cut is the total weight of the remaining edge between the last two nodes.

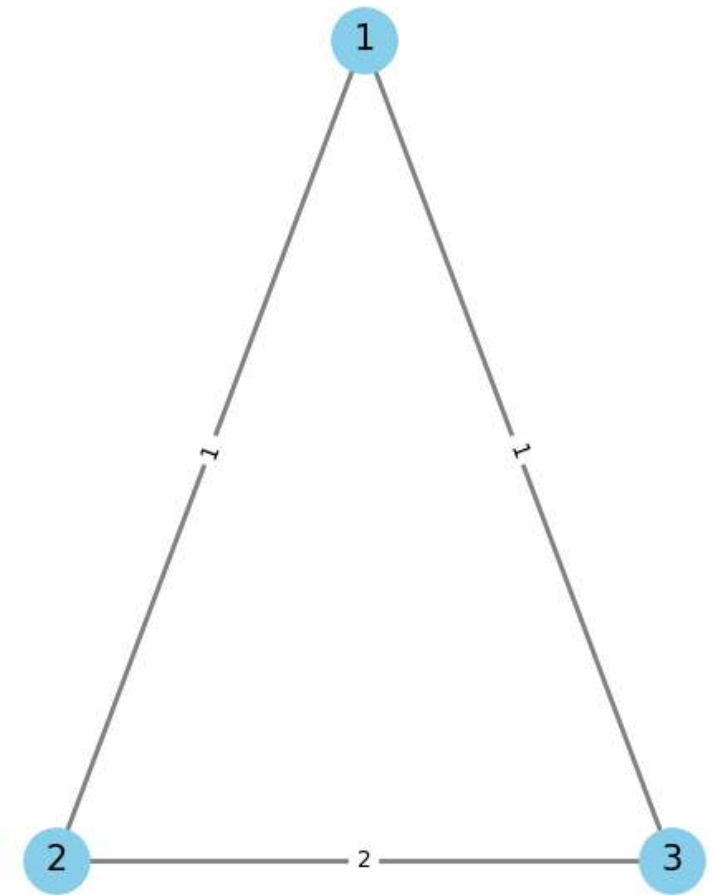
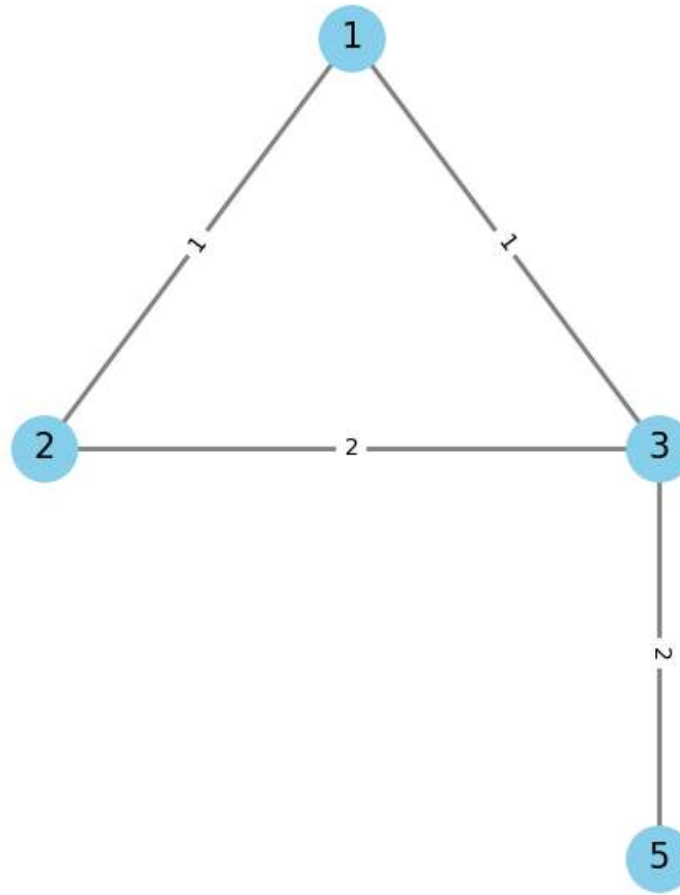


KARGER'S ALGORITHM

Iteration 1: Graph after merged 3 and 4

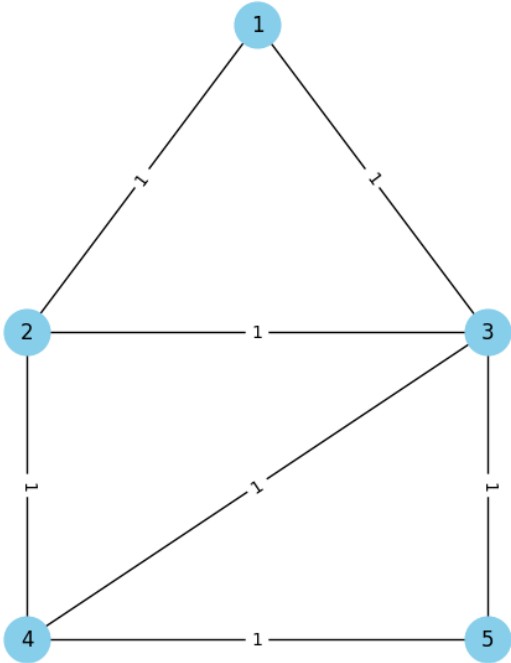


Iteration 2: Graph after merged 3 and 5

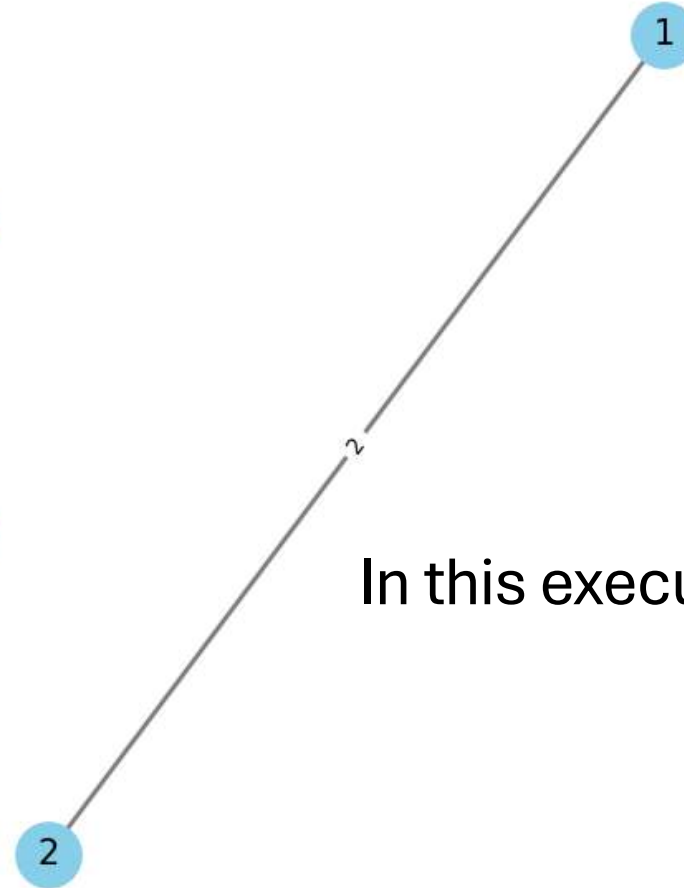


KARGER'S ALGORITHM

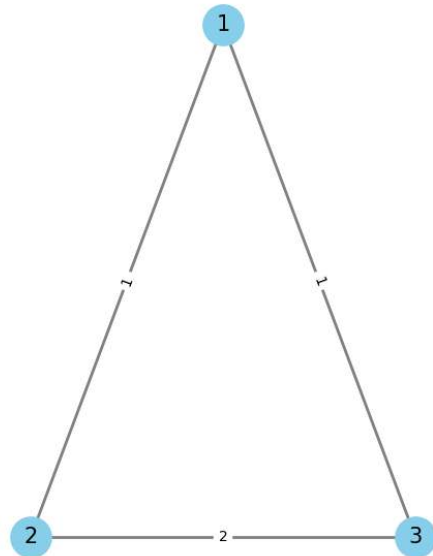
Iteration 1: Graph after merged 3 and 4



Iteration 3: Graph after merged 2 and 3



Iteration 2: Graph after merged 3 and 5



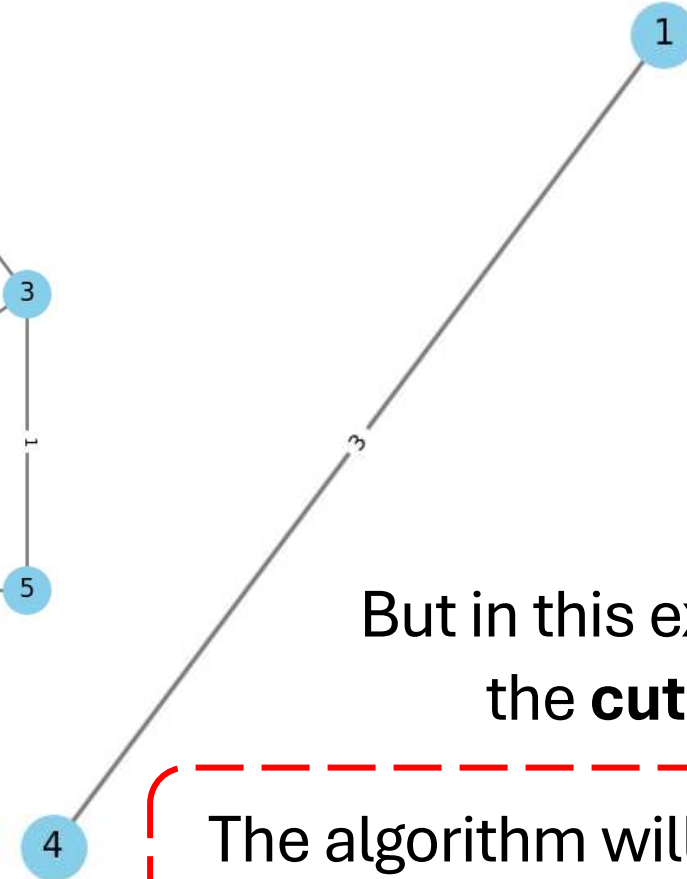
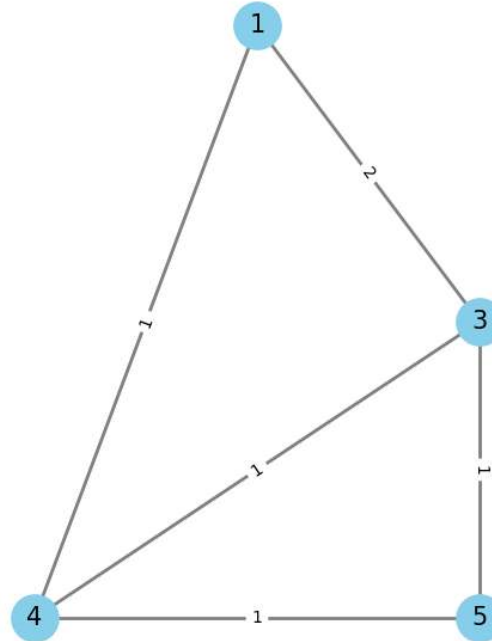
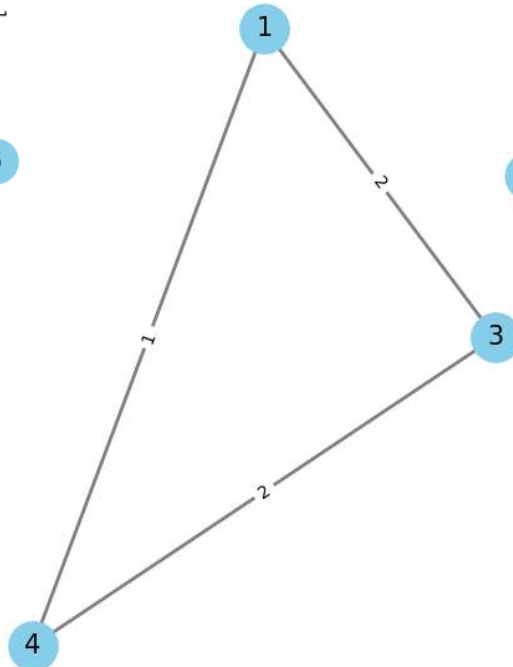
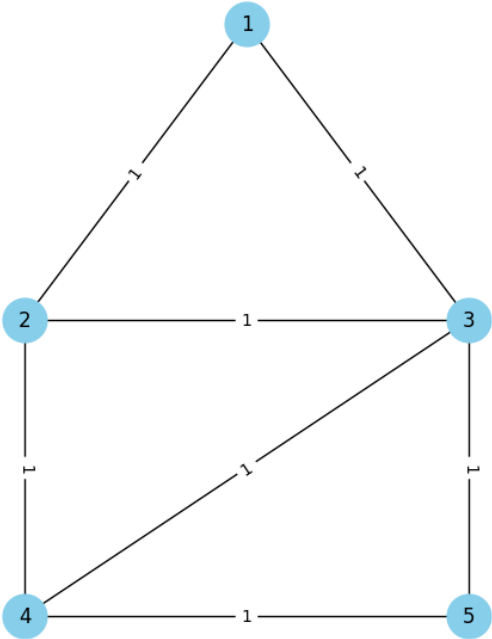
In this execution the **cut** is 2

KARGER'S ALGORITHM

Iteration 1: Graph after merged 1 and 2

Iteration 3: Graph after merged 1 and 3

Iteration 2: Graph after merged 4 and 5

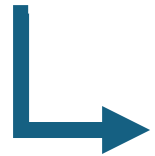


But in this execution
the **cut** is 3

The algorithm will not always
return 2.

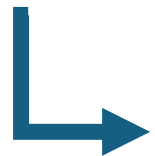
KARGER'S ALGORITHM

To increase the probability of finding the true minimum cut, the algorithm must be executed multiple times and take the minimum of the results.



In a graph with n nodes, each run has the chance to produce the minimum cut with probability of:

$$\frac{2}{n(n-1)}$$



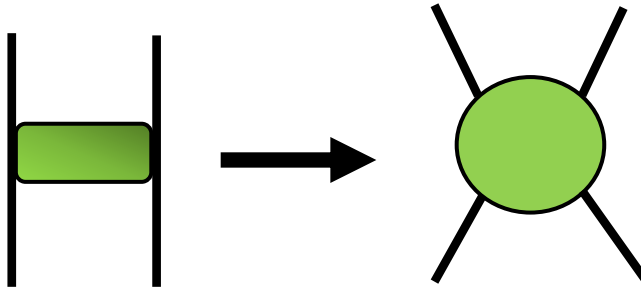
The total runtime of the algorithm is $O(n^4)$.

KARGER'S ALGORITHM

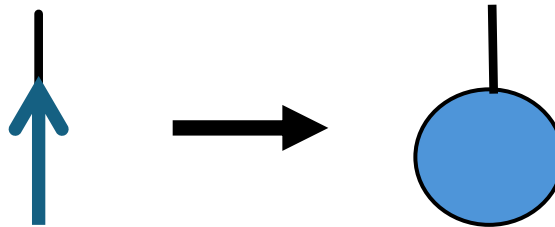
 Implementing the algorithm on a quantum circuit.

1. The quantum circuit must be converted to a graph.

Two-Qubits gates:



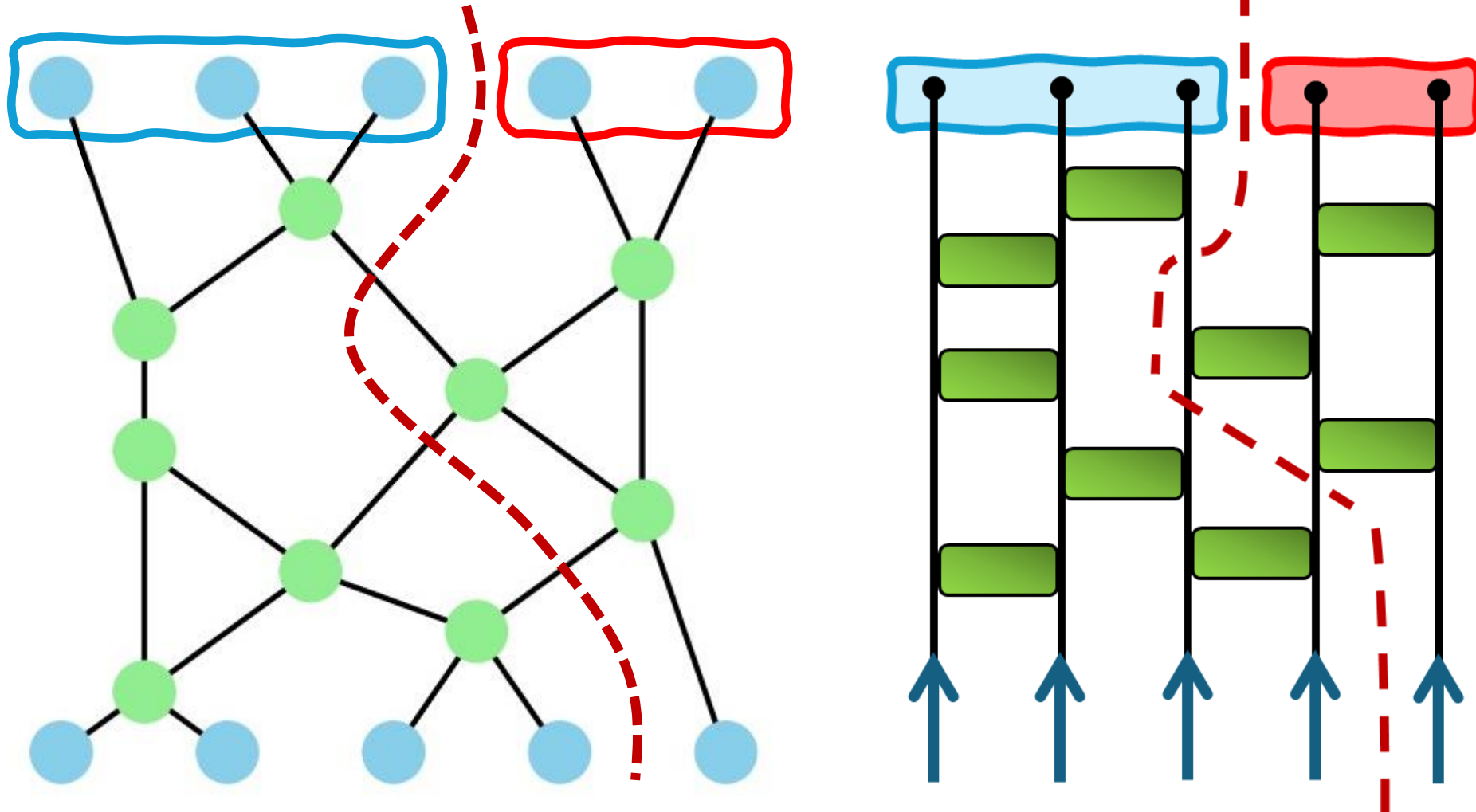
Qubits:



KARGER'S ALGORITHM

❏ Implementing the algorithm on a quantum circuit.

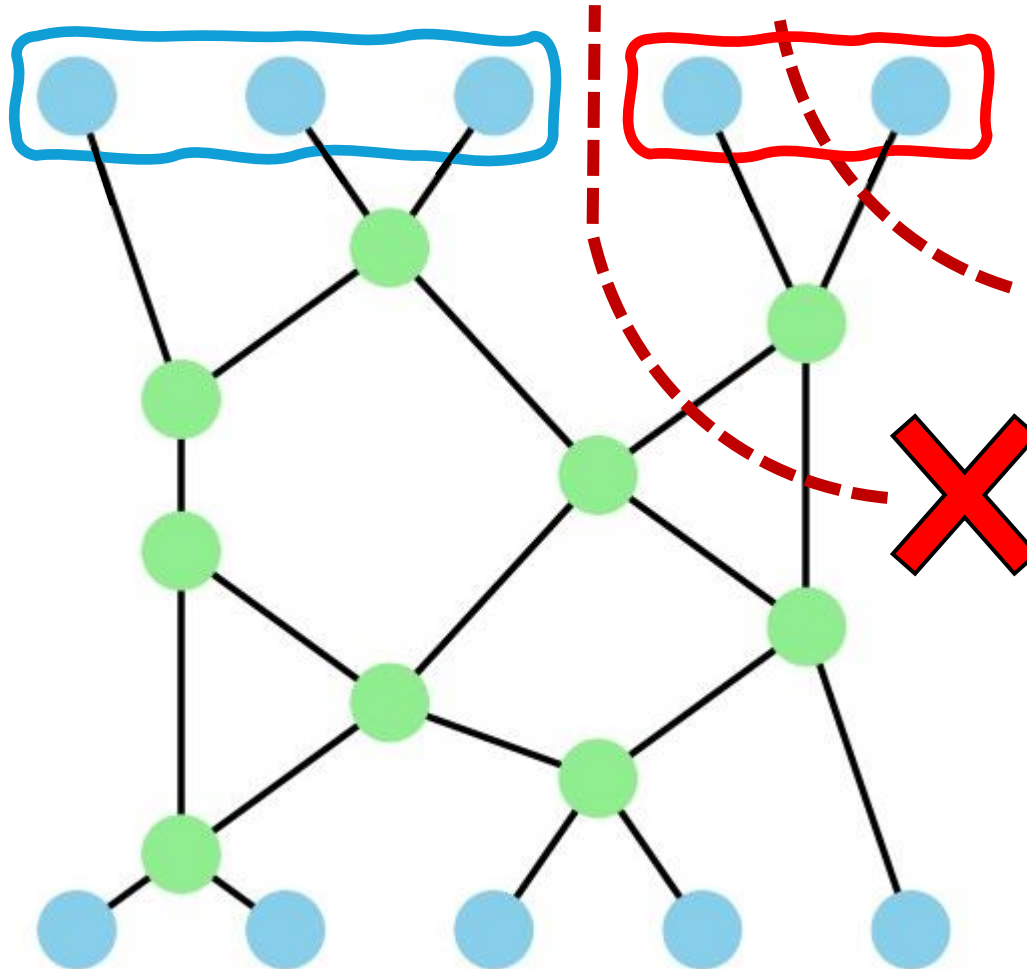
1. The quantum circuit must be converted to a graph.



KARGER'S ALGORITHM

📦 Implementing the algorithm on a quantum circuit is **not immediate**.

➡ The minimum cut obtained by Karger's algorithm, in this graph, will be 1.



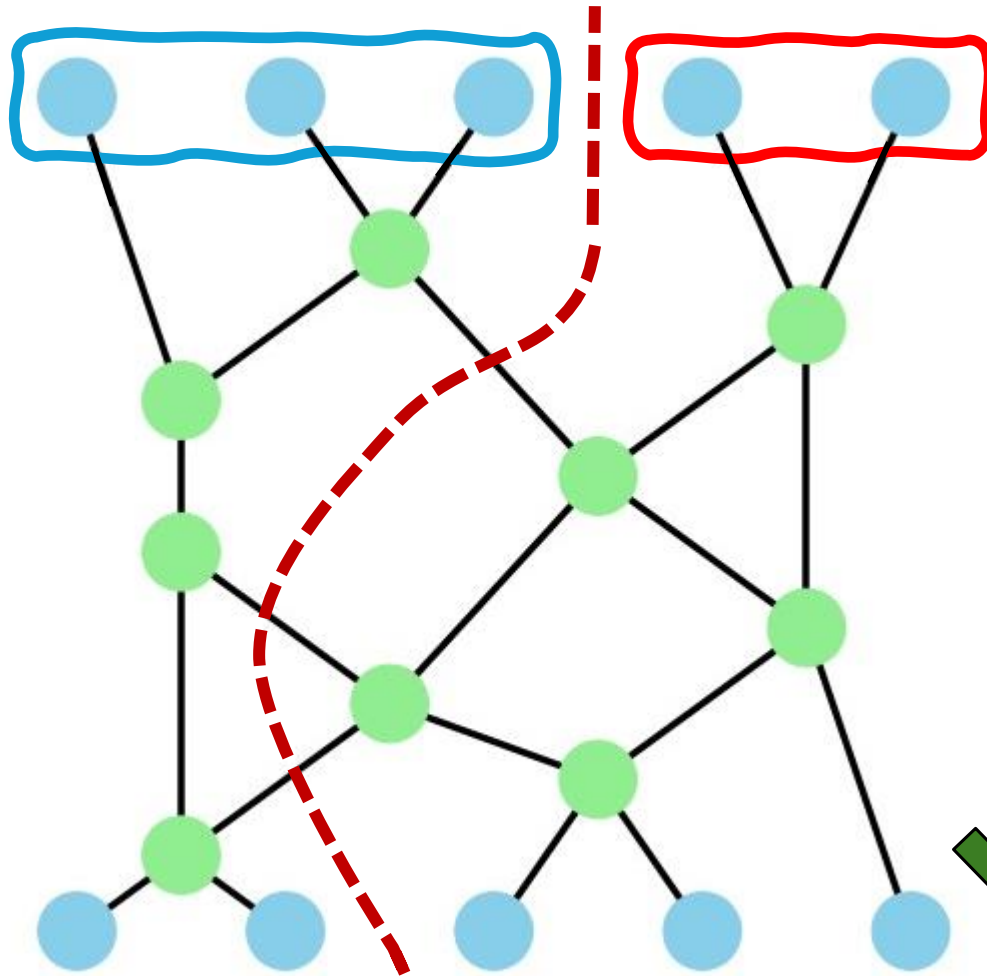
Restrictions

1. The cut must be made through the middle of the two subsystems.
2. The cut must come out from below the circuit.

KARGER'S ALGORITHM

❏ Implementing the algorithm on a quantum circuit is **not immediate**.

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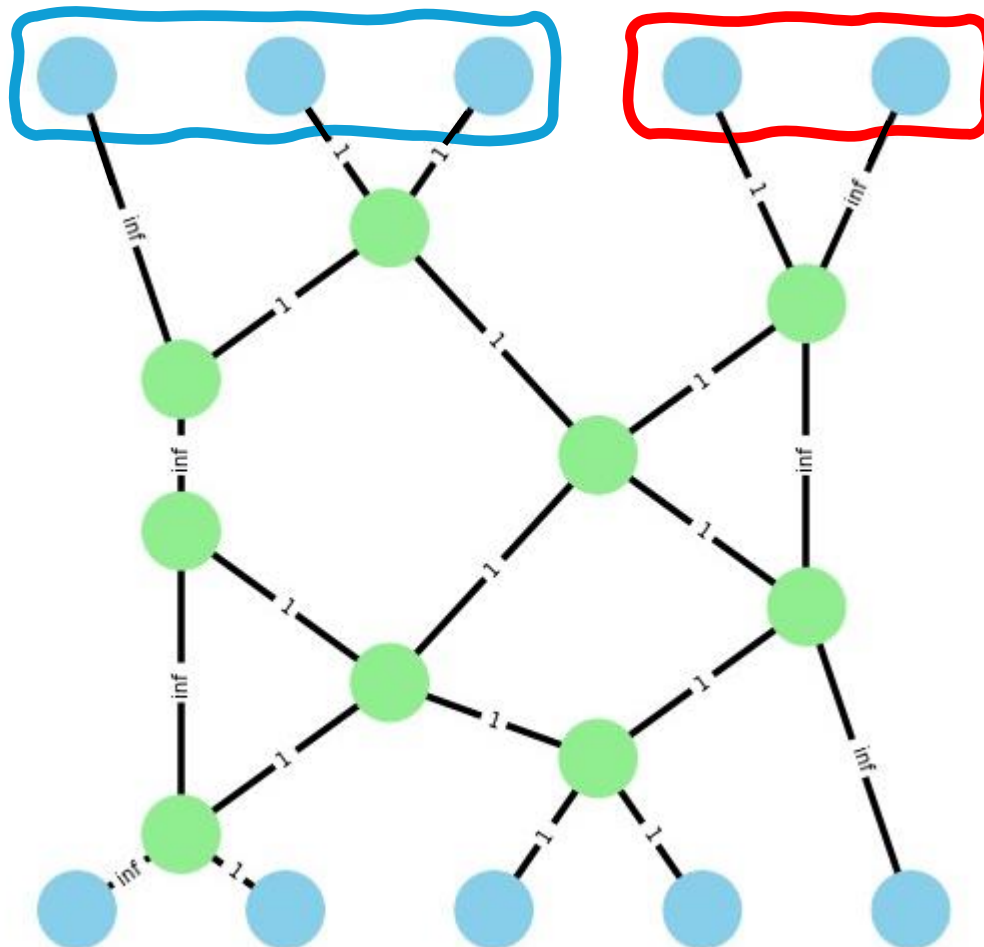
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KARGER'S ALGORITHM

❏ Implementing the algorithm on a quantum circuit is **not immediate**.

2. The graph needs to be modified.



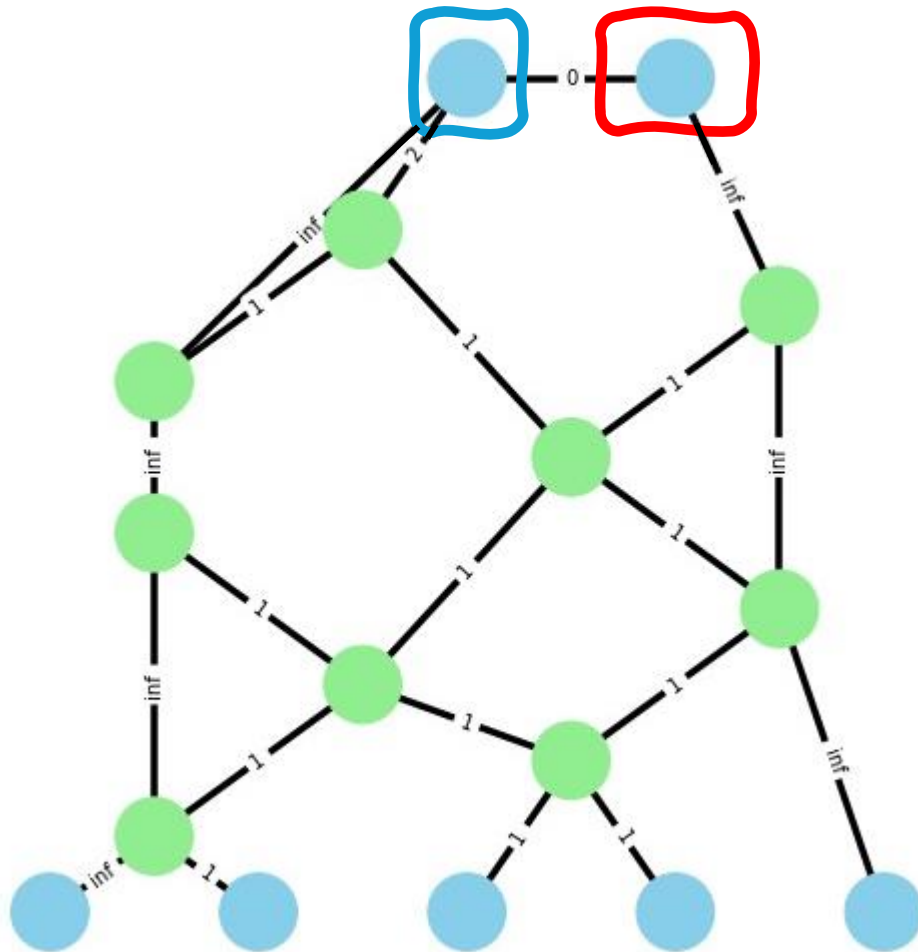
Restrictions

1. Assign a large weight to the border edges.

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❏ Implementing the algorithm on a quantum circuit is **not immediate**.

2. The graph needs to be modified.

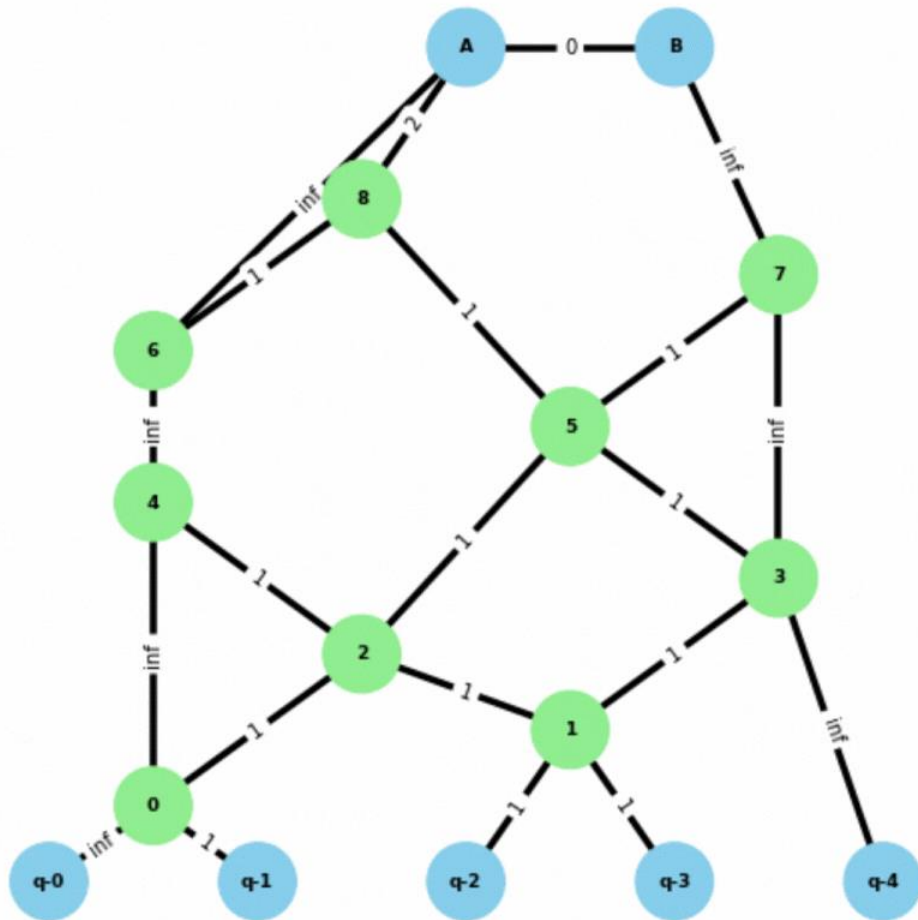


Restrictions

1. Assign a large weight to the border edges.
2. Merge all the top qubits (nodes) of each subsystem into two unique nodes, and put an edge between them with weight 0.
3. The last two nodes that survive in the Karger algorithm will be the two top nodes (the two subsystems)

KARGER'S ALGORITHM

❏ Implementing the algorithm on a quantum circuit is **not immediate**.



In a circuit with **N qubits** and **L gates**, we will have $n = N + L + 2$ nodes.



If we run the algorithm $\frac{n(n-1)}{2} \cdot P\%$ times differently, we ensure the minimum cut with $P\%$ probability.

RANDOM UNITARY CIRCUITS

-  A random circuit is one for which either **the structure of operators in time in space**, or the **operators themselves**, are chosen in some random way.

Why study random circuits?

- ☐ There is the hope that entanglement dynamics under a random circuit will be generic in some way.
- ☐ Studying a specific nonintegrable model is hard. But in the random circuit it is often possible to obtain exact results for averages over the ensemble.
- ☐ They are used to study the propagation of quantum information and the emergence of chaos in many-body quantum systems.

RANDOM UNITARY CIRCUITS

-  A random circuit is one for which either **the structure of operators in time in space**, or the **operators themselves**, are chosen in some random way.

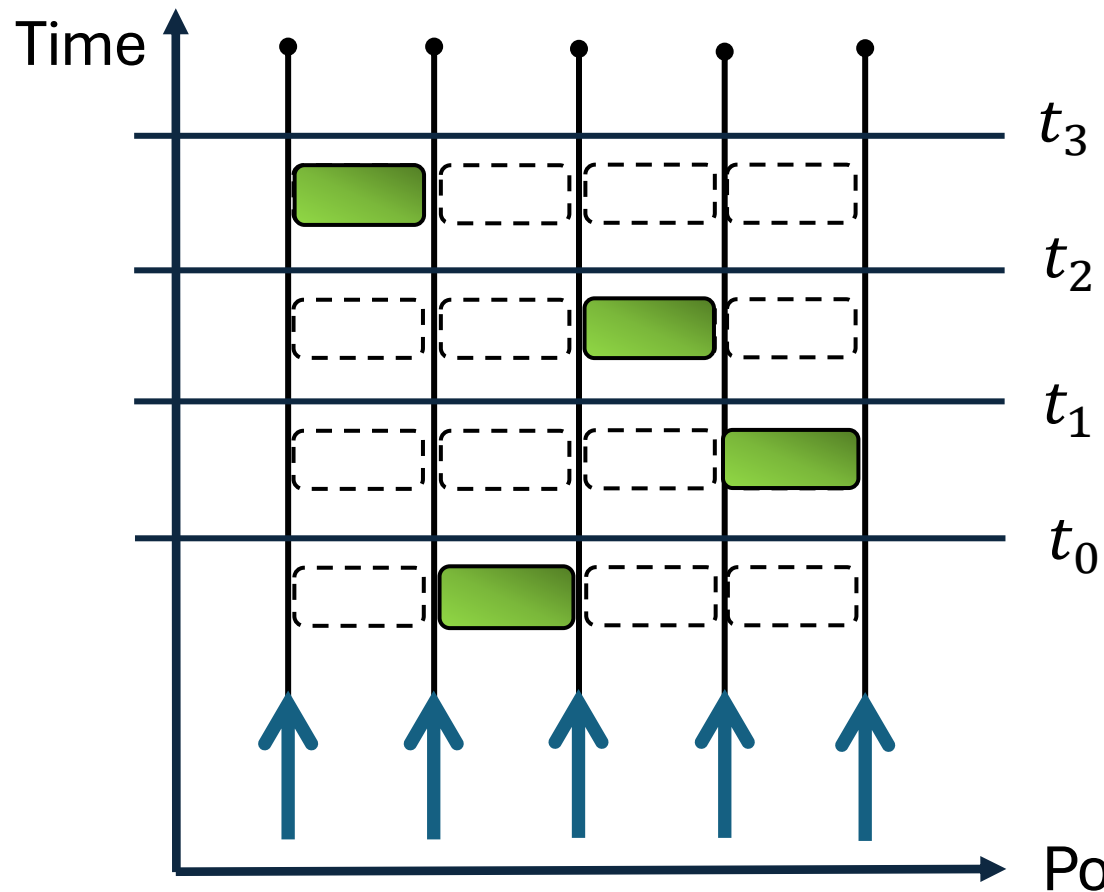
Why study random circuits?



This is in the general spirit of other uses of randomness such as random matrix theory.

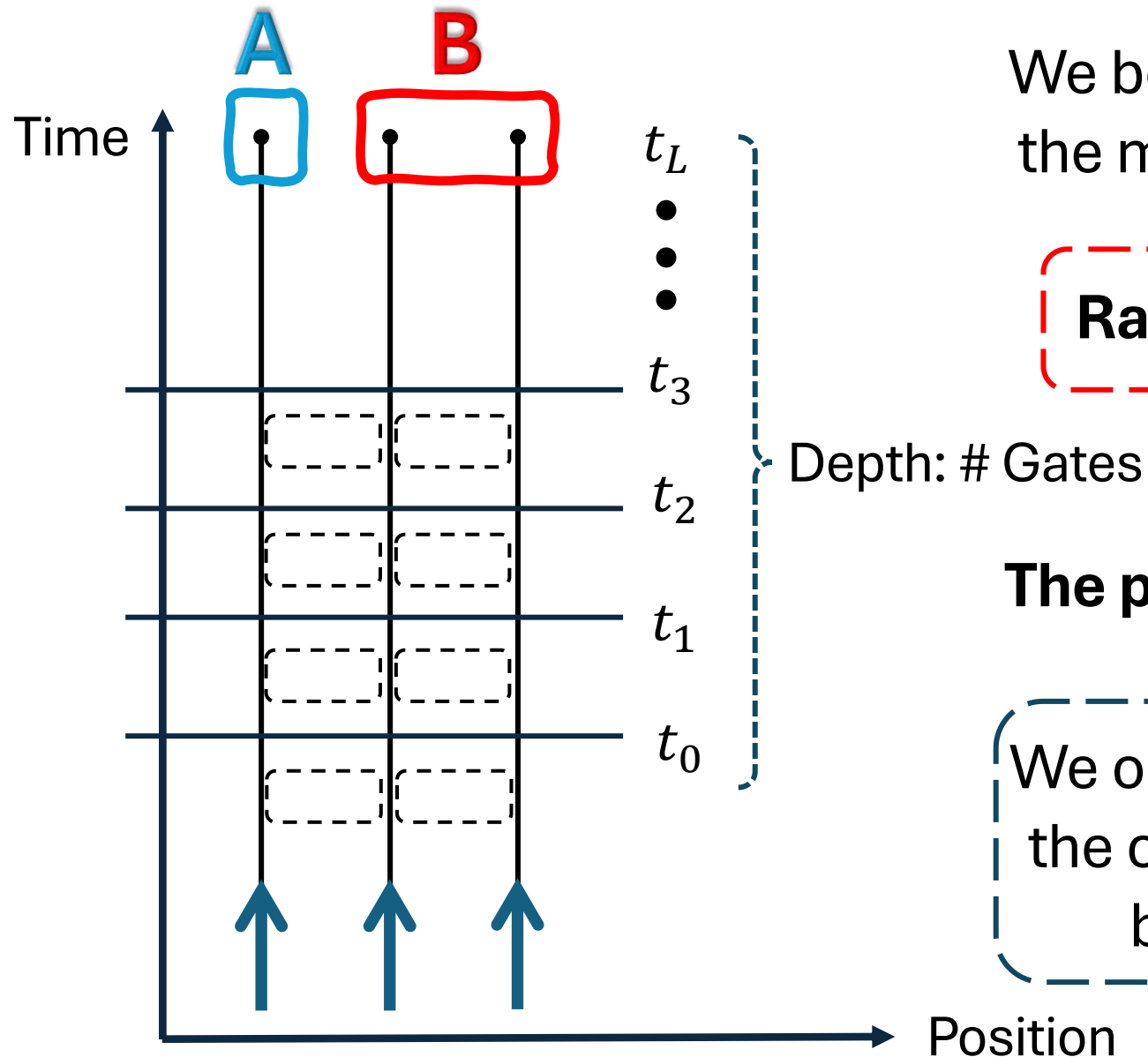
STUDY OF TWO-QUBITS GATE WITH UNIFORM PROBABILITY

The objective is to study circuits with random architecture of **local two-qubit gates** (two neighboring qubits) with a uniform probability distribution.



At time t a gate can be placed in any position with equal probability

STUDY OF TWO-QUBITS GATE WITH UNIFORM PROBABILITY



We begin by studying the behavior of the minimum cut in a simple circuit.

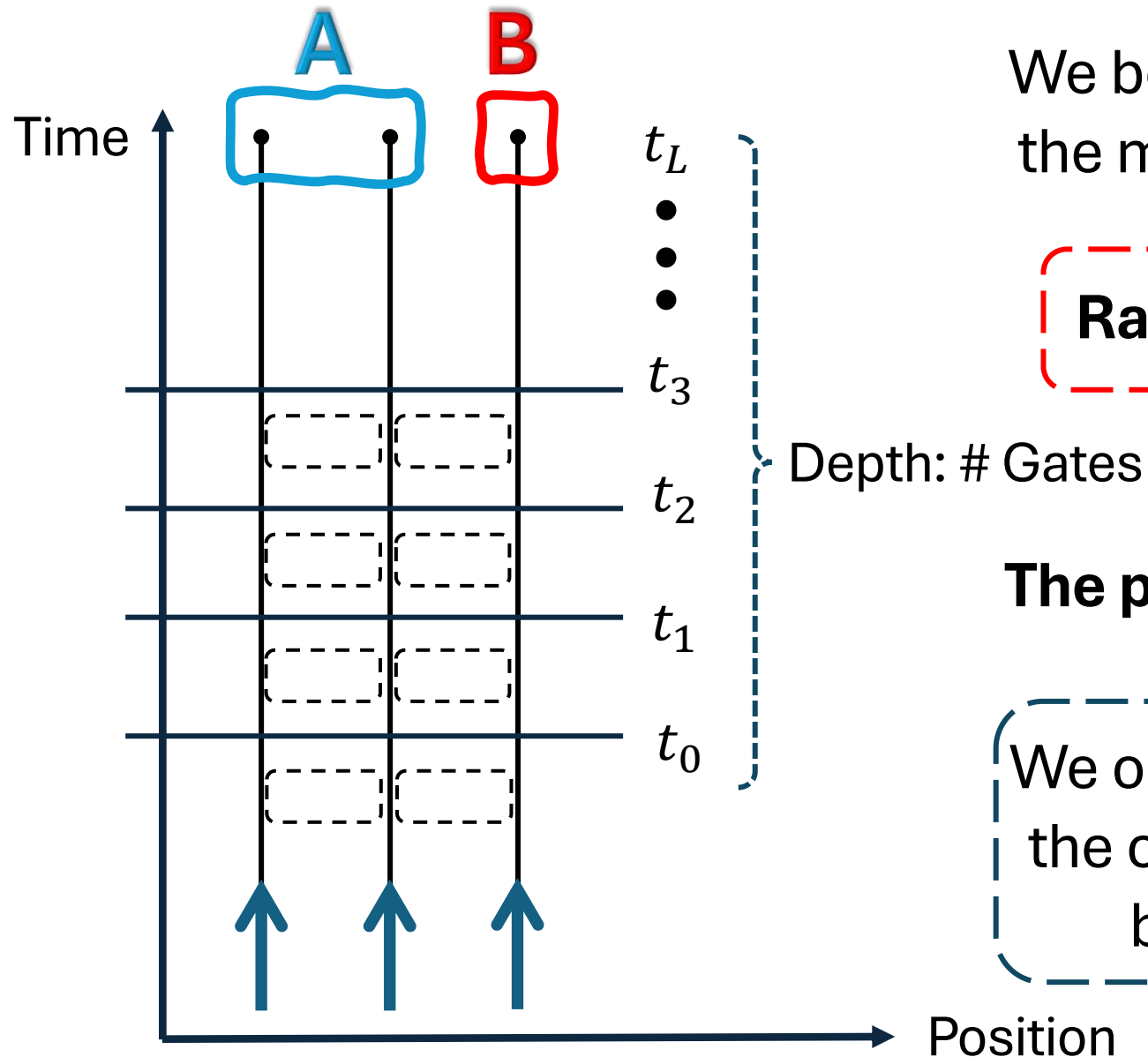
Random circuits with 3 qubits



The problem is analytically solvable

We only have two options to separate the circuit into two subsystems, and both cases are symmetrical.

STUDY OF TWO-QUBITS GATE WITH UNIFORM PROBABILITY



We begin by studying the behavior of the minimum cut in a simple circuit.

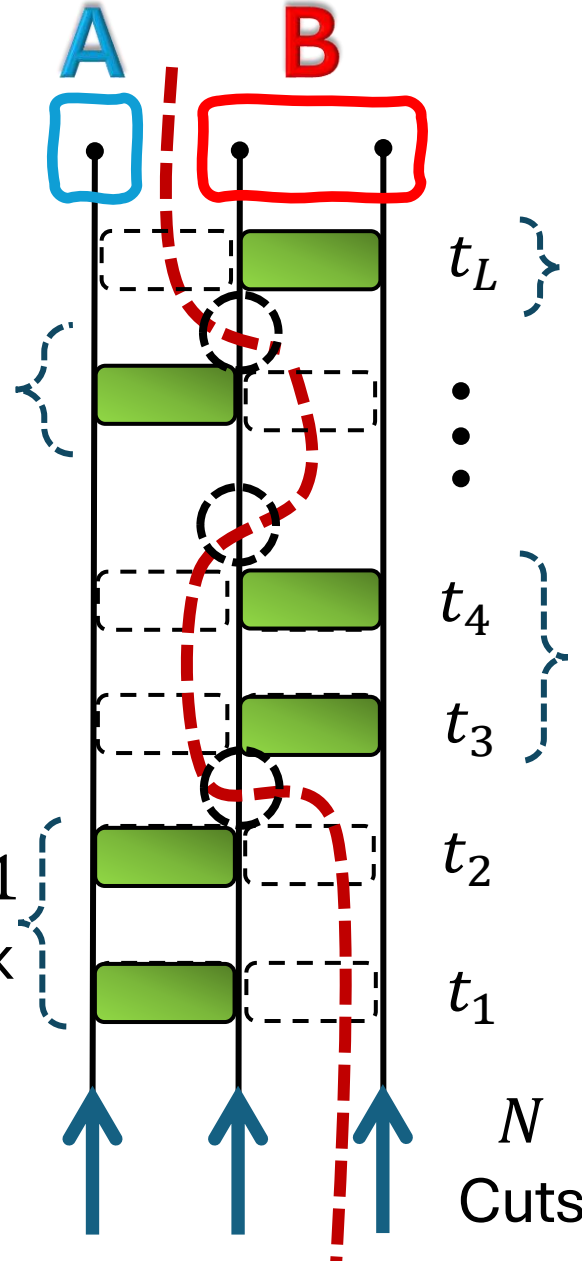
Random circuits with 3 qubits



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STUDY OF TWO-QUBITS GATE WITH UNIFORM PROBABILITY



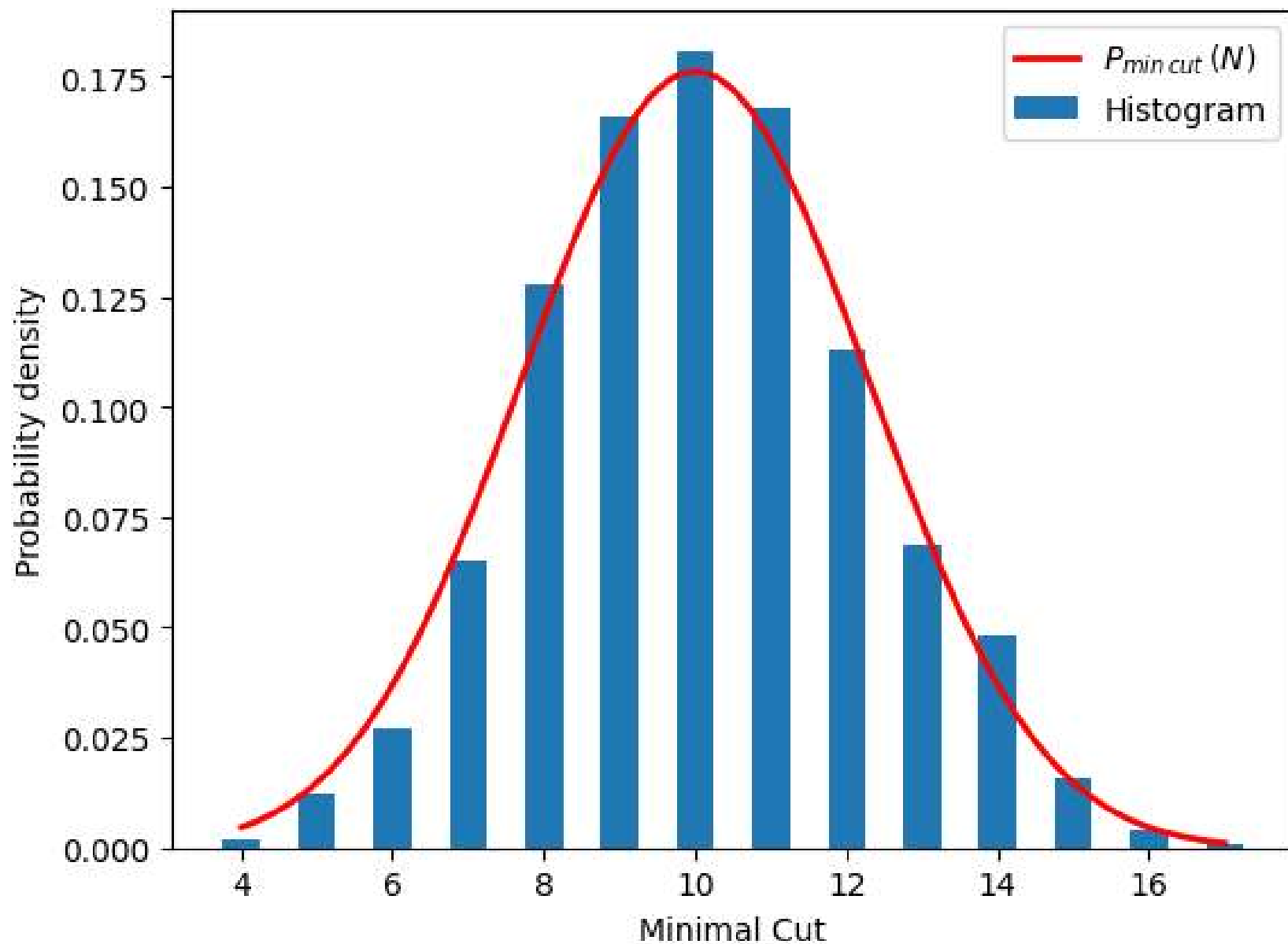
Random circuits with 3 qubits

- There will be a cut for each obstacle (Block) in the circuit.
- Since we have two possible positions, distributing L gates in $N + 1$ blocks becomes a binomial distribution.

$$P_{\text{min cut}}(N) = \frac{\# \text{ Circuits with } N+1 \text{ blocks}}{\# \text{ Total circuits}}$$

$$P_{\text{min cut}}(N) = \frac{1}{2^L} \binom{L}{N} = \frac{L!}{2^L N! (L-N)!}$$

RANDOM CIRCUITS WITH 3 QUBITS

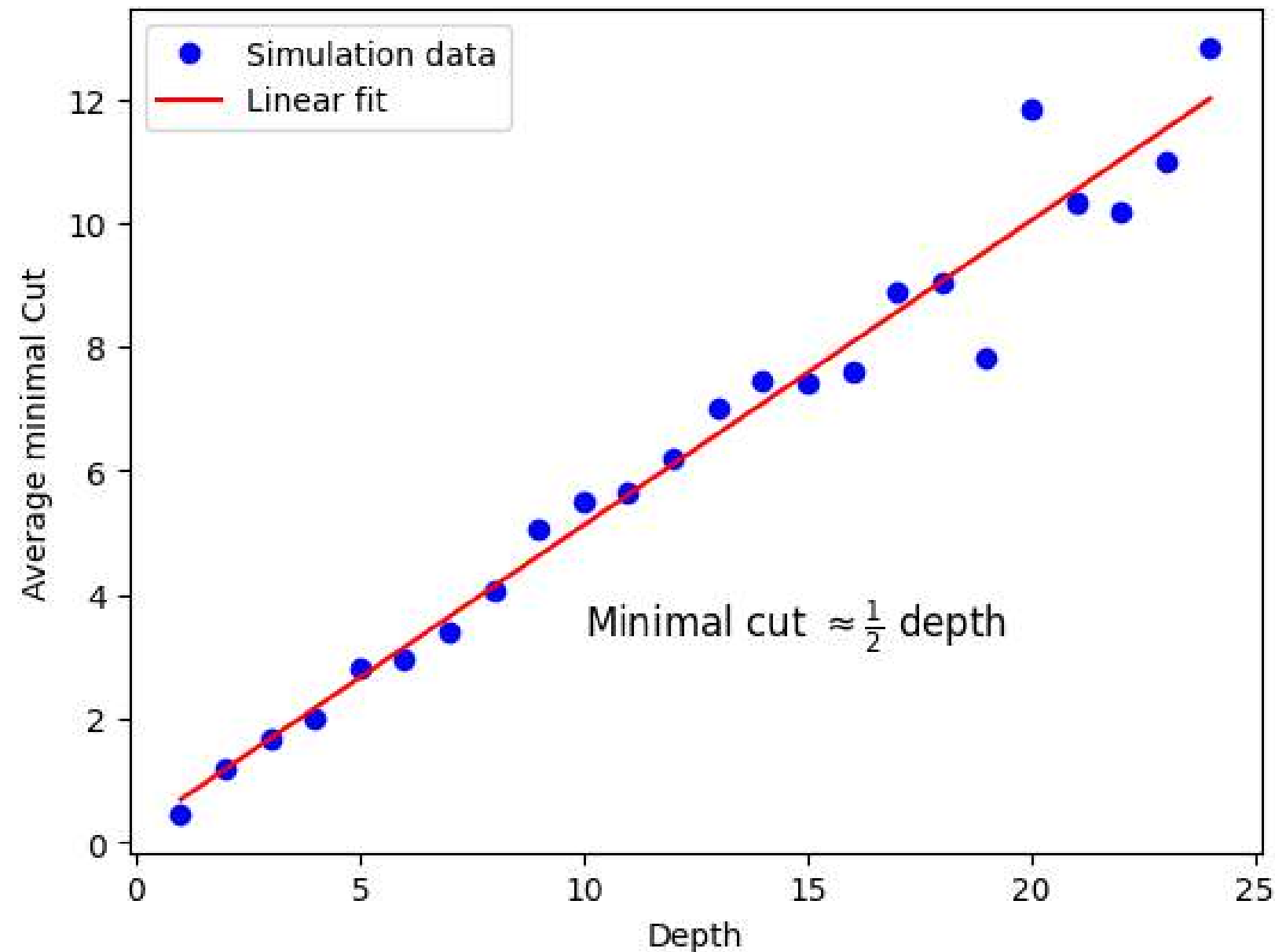


□ Results of implementing the Karger's algorithm on 1000 random circuits with 20 gates

Note that there are $(\text{qubits} - 1)^{\text{Gates}}$ possible circuits in total.

$$2^{20} = 1,048,576$$

RANDOM CIRCUITS WITH 3 QUBITS



□ Experiments with 600 random circuits for each number of gates

$$600 \ll 2^L, L > 20$$

RANDOM CIRCUITS WITH 3 QUBITS

- ❑ In a random circuit with L gates the maximum **minimal cut** is L

↳ Entropy cannot exceed L : $S_n \leq L$

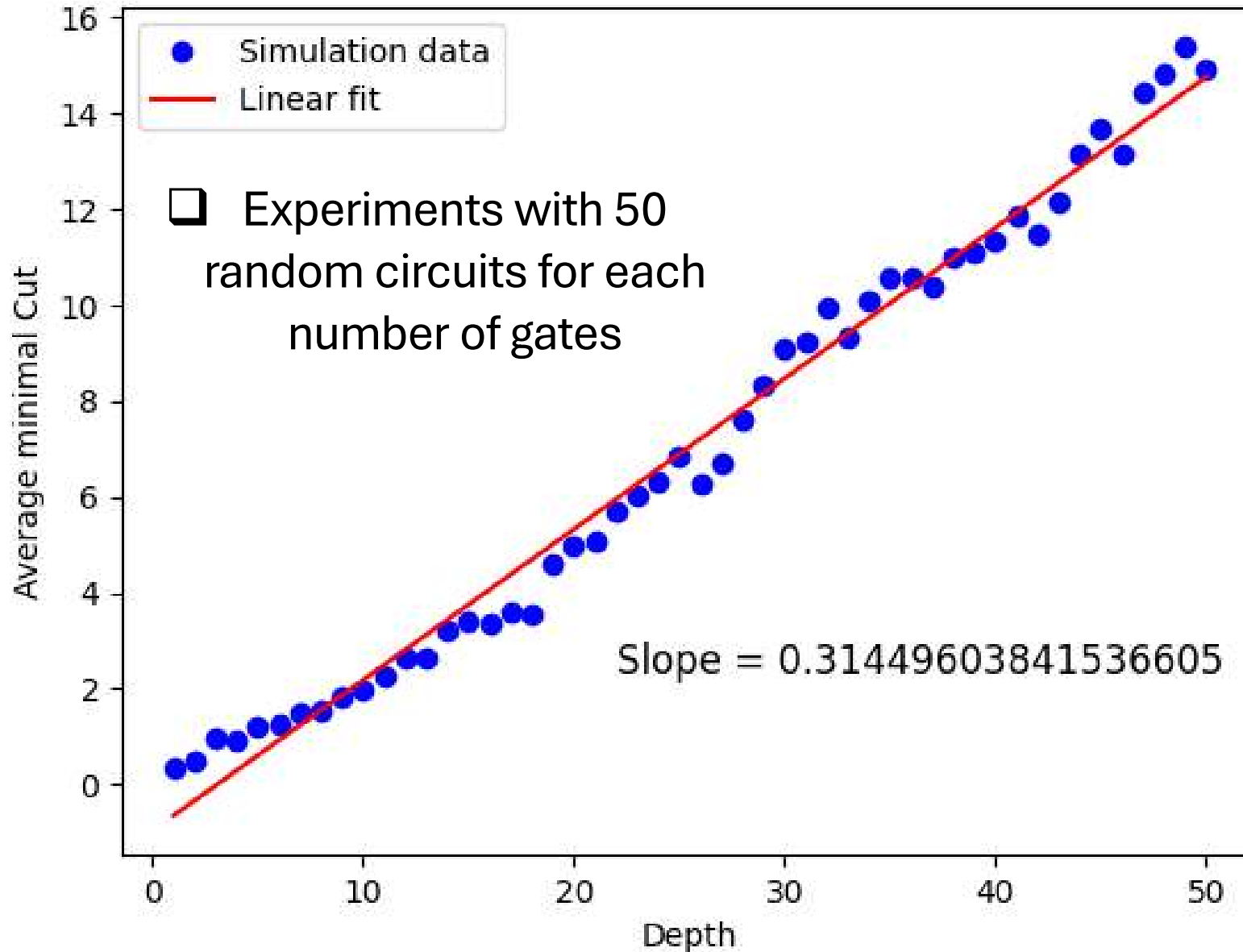
- ❑ The most probable **minimal cut** is $\frac{L}{2}$

↳ There is more than 50% probability that the **entropy** is : $S_n \leq \frac{L}{2}$

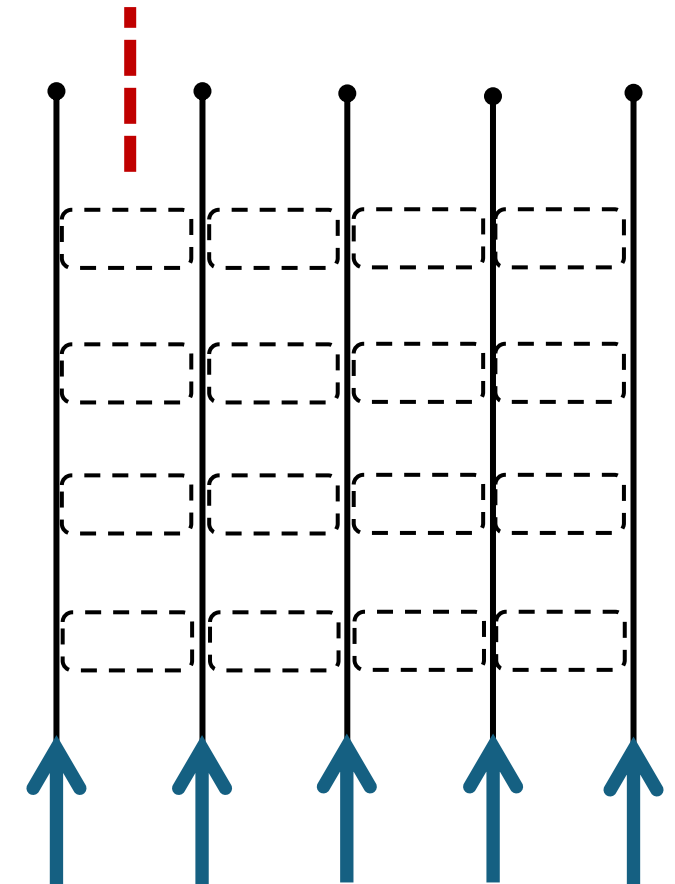
RANDOM CIRCUITS WITH MORE QUBITS?

PRELIMINARY APPROACH AND INITIAL RESULTS

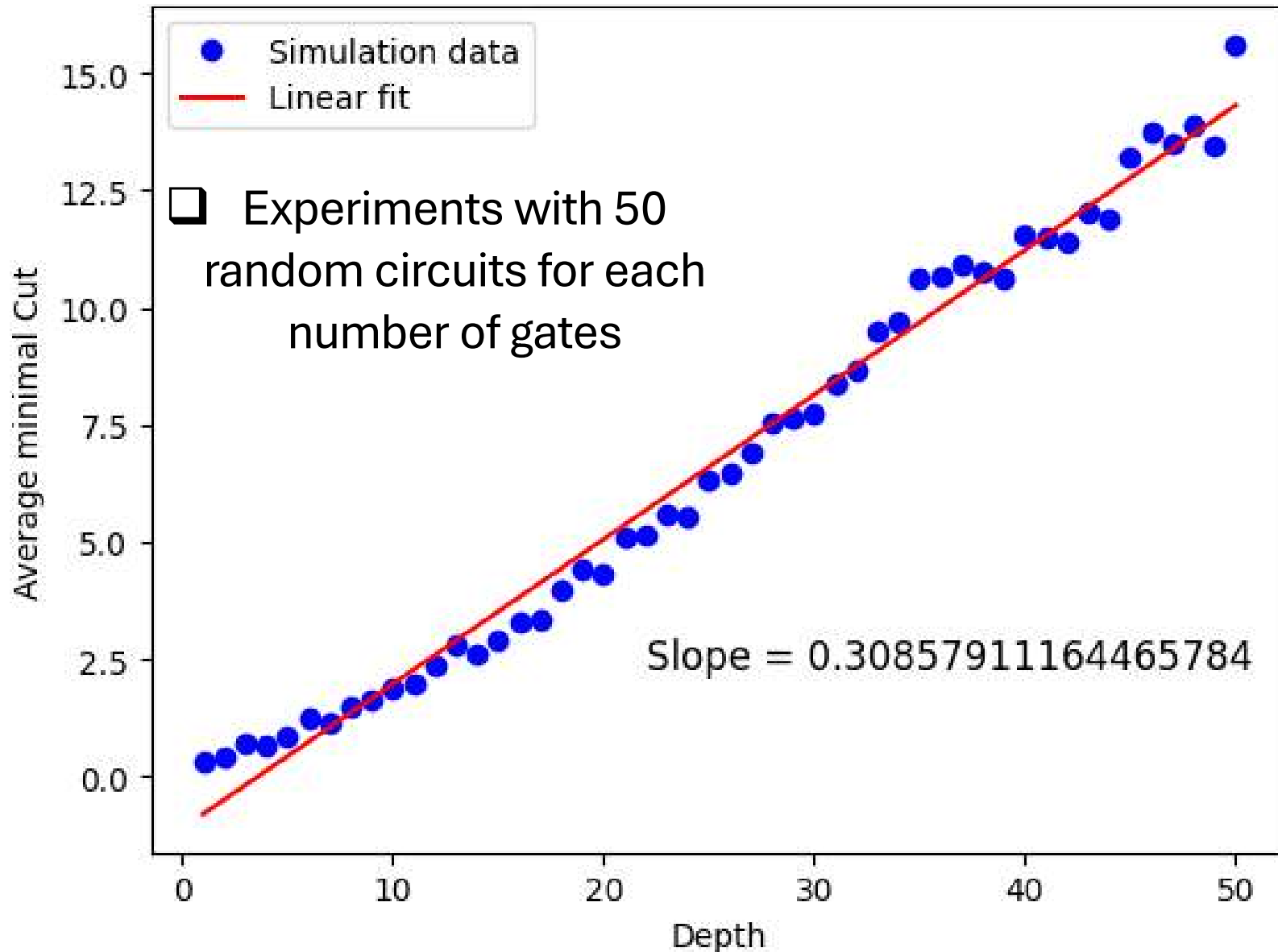
STUDY OF TWO-QUBITS GATE WITH UNIFORM PROBABILITY



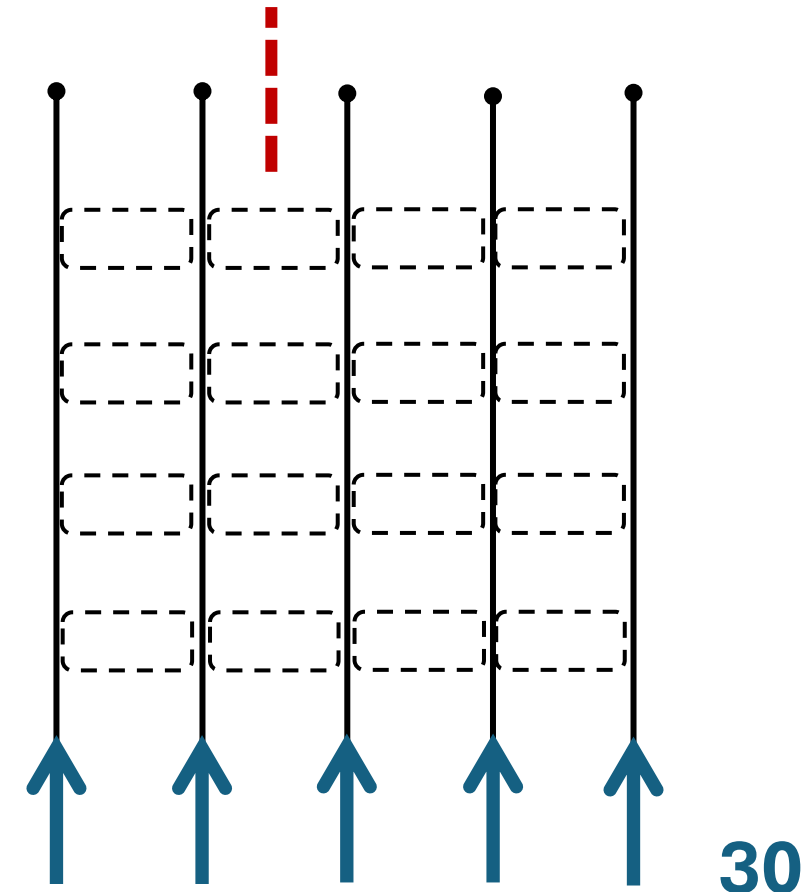
Quantum circuit with **5 qubits**, minimal cut passing through qubit 1 and 2



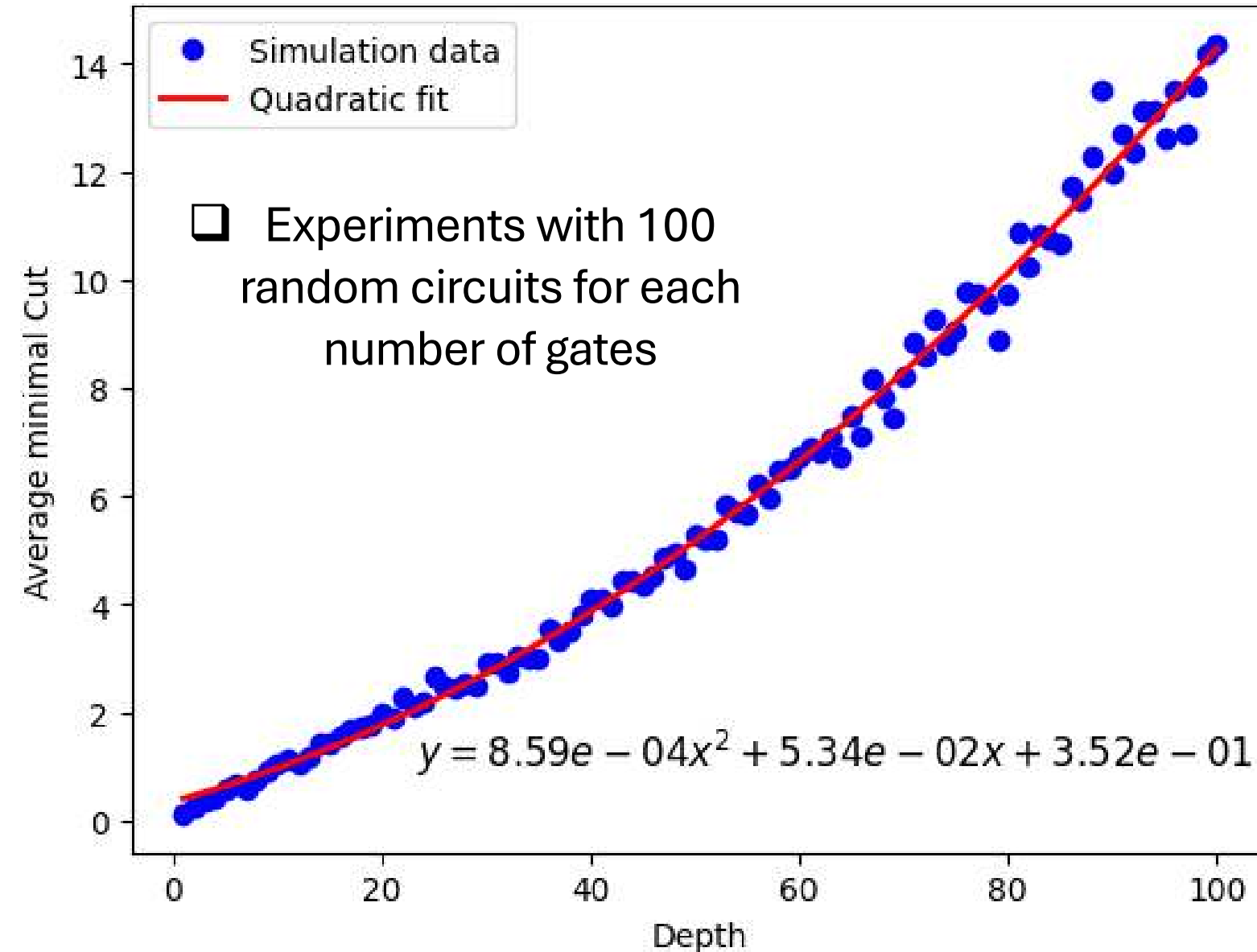
STUDY OF TWO-QUBITS GATE WITH UNIFORM PROBABILITY



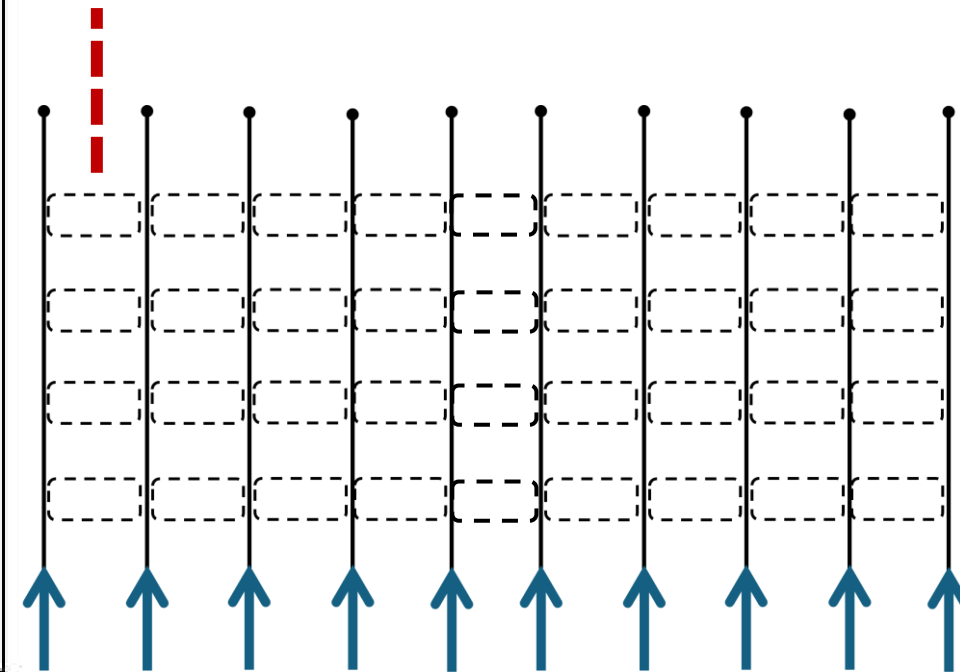
Quantum circuit with **5 qubits**, minimal cut passing through qubit 2 and 3



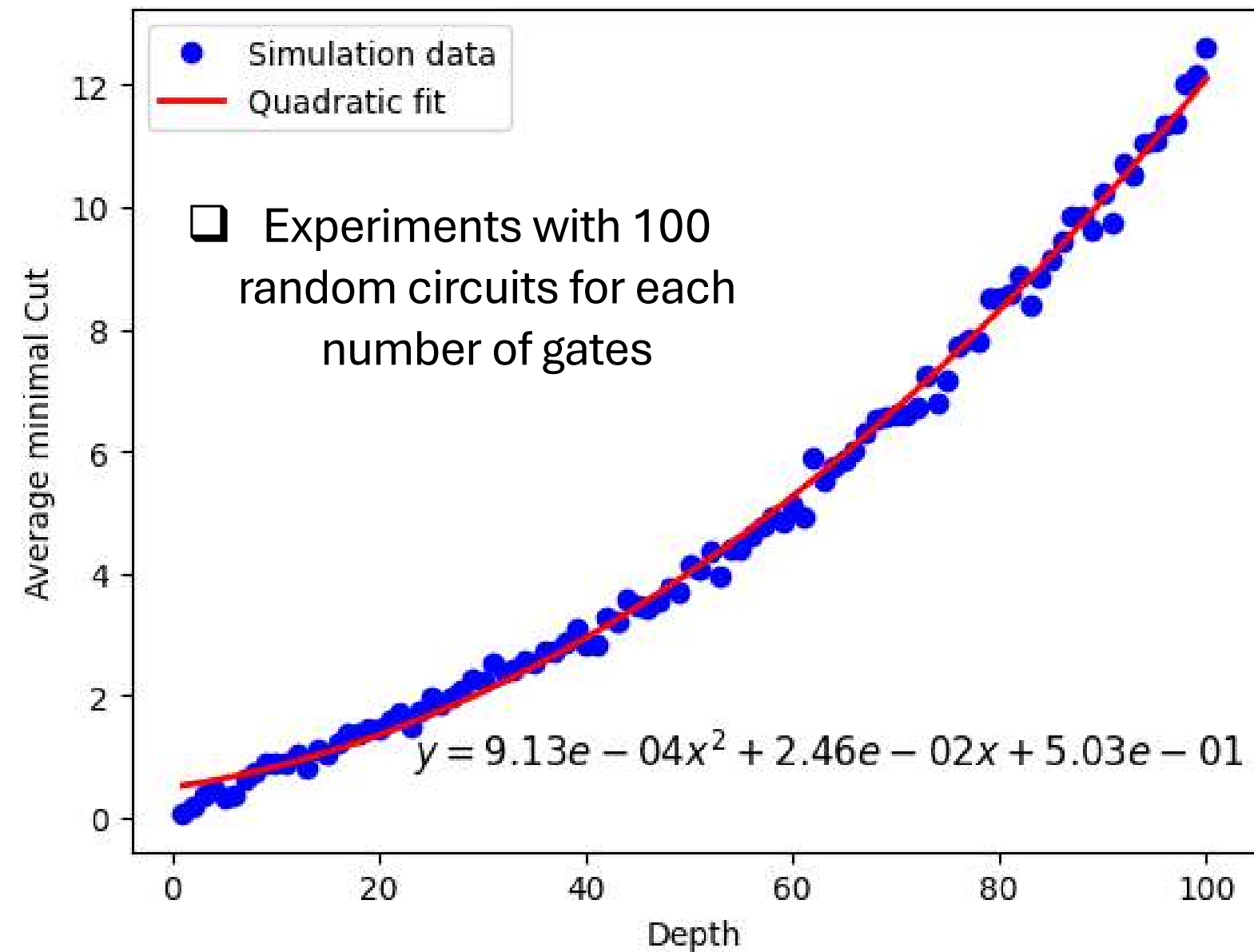
STUDY OF TWO-QUBITS GATE WITH UNIFORM PROBABILITY



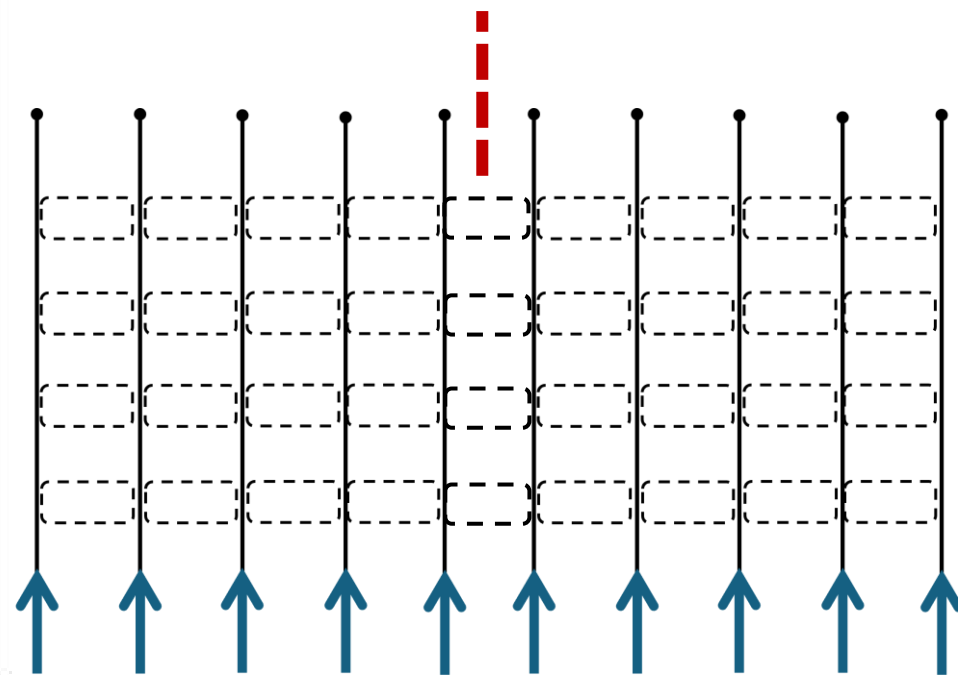
Quantum circuit with **10 qubits**, minimal cut passing through qubit 1 and 2



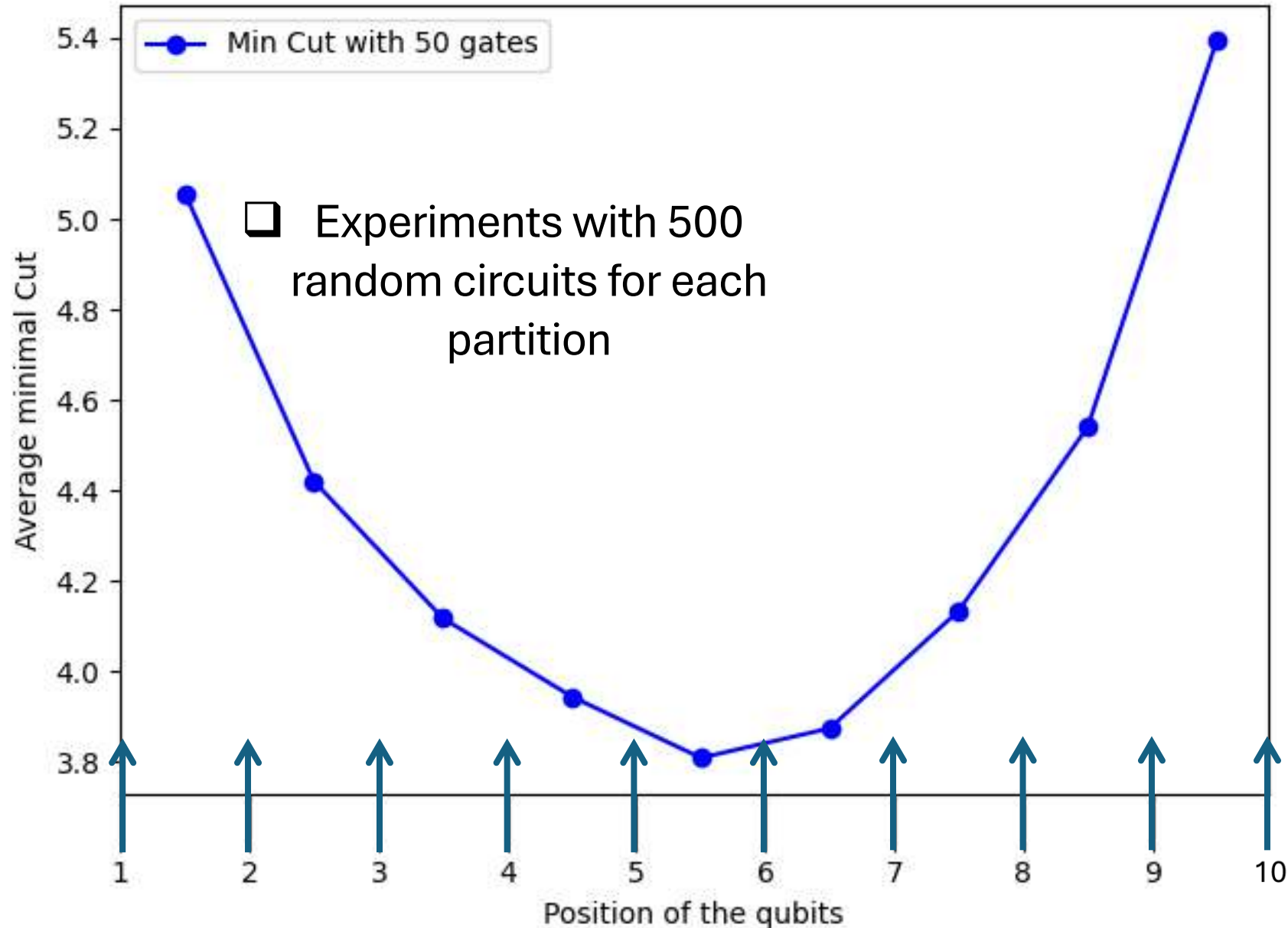
STUDY OF TWO-QUBITS GATE WITH UNIFORM PROBABILITY



Quantum circuit with **10 qubits**, minimal cut passing through qubit 5 and 6

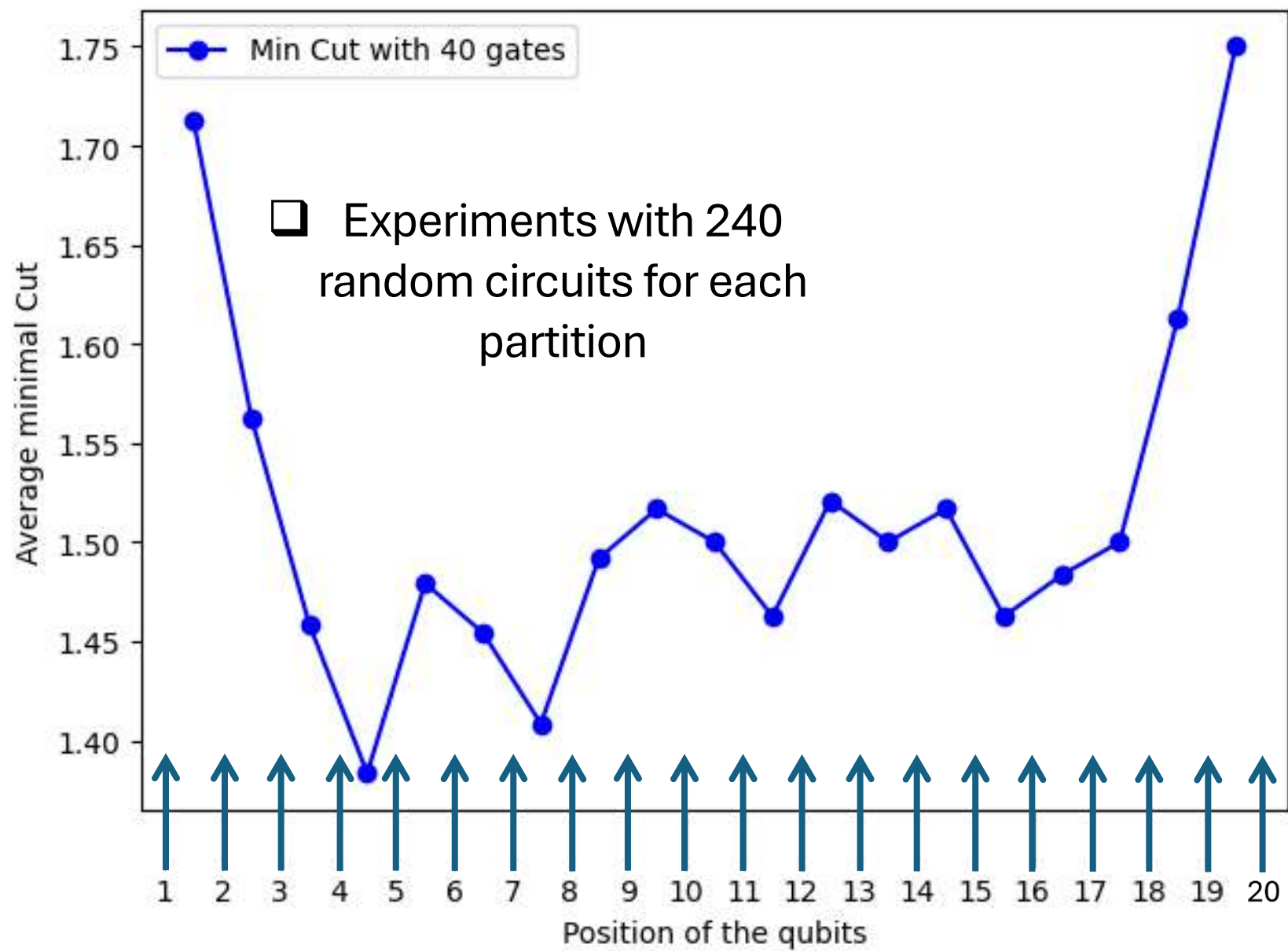


STUDY OF TWO-QUBITS GATE WITH UNIFORM PROBABILITY



Minimal cut that passes through each intersection of two qubits (partition) of a random circuit with **10 qubits** and **50 gates**.

STUDY OF TWO-QUBITS GATE WITH UNIFORM PROBABILITY



Random circuit with **20 qubits** and **40 gates**.

There are more possibilities of finding a smaller minimal cut through the centers than through the borders.

CONCLUSIONS

❑ The minimal cut grows with the number of gate.

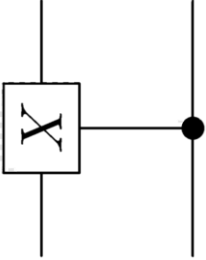
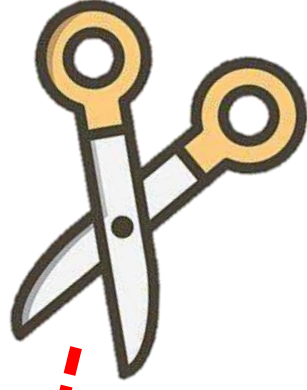
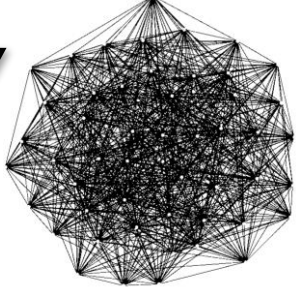
↳ Entropy has more freedom to grow in circuits with many gates.

❑ The minimal cut grows more at the borders of the circuit, than in partitions through the center.

↳ The border qubits have more freedom of entanglement entropy with the rest of the circuit.

PERSPECTIVES

- ❑ Implement more optimal minimal cut algorithms.
- ❑ Calculating exact entropies and making comparisons.
- ❑ Obtaining analytical results in the thermodynamic limit of the number of qubits and circuit depth.
- ❑ Exploring other probability distributions.

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